MILITARY SPECIFICATION

MICROCIRCUITS, MONOLITHIC SILICON INTERFACE, DUAL PERIPHERAL DRIVERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for bipolar, monolithic silicon, dual peripheral drivers. Two product assurance classes and a choice of case outline/lead finish are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual NAND gate and transistor (separate), high speed switching
02	Dual AND gate and transistor (connected) high speed switching
03	Dual NAND gate and transistor (connected), high speed switching
04	Dual OR gate and transistor (connected), high speed switching
05	Dual NOR gate and transistor (connected), high speed switching
06	High voltage dual NAND gate and transistor (separate), medium speed switching
07	High voltage dual AND gate and transistor (connected), medium speed switching
08	High voltage dual NAND gate and transistor (connected), medium speed switching
09	High voltage dual OR gate and transistor (connected), medium speed switching
10	High voltage dual NOR gate and transistor (connected), medium speed switching

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
P	D-4 (8-lead, 1/4" x 3/8"), dual-in-line package

1.3 Absolute maximum ratings.

Supply voltage (V_{CC}) - - - - - - - - - - - - - 7 V dc

Input voltage (V_{IN})- - - - - - - - - - - - - - 5.5 V dc

Interemitter voltage (V_{EM}) - - - - - - - - - - 5.5 V dc

V_{CC} to substrate voltage (V_{CCS}):

Device type 01 - - - - - - - - - - - - - - - 35 V dc
Device type 06 - - - - - - - - - - - - - - - 40 V dc

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings (continued).

Collector to substrate voltage (V_{CS}):	
Device type 01	35 V dc
Device type 06	40 V dc
Collector to base voltage (V_{EB}):	
Device type 01	35 V dc
Device type 06	40 V dc
Collector to emitter voltage (V_{CER}):	
Device type 01	30 V dc
Device type 06	40 V dc
Collector to emitter voltage (V_{CEO}):	
Device type 06	25 V dc
Emitter to base voltage (V_{BE}):	
Device types 01 and 06	5 V dc
Offstate output voltage (V_{OO}):	
Device types 02 through 05	30 V dc
Device types 07 through 10	35 V dc
Continuous collector current (I_{CC}):	
Device types 01 and 06	300 mA
Continuous output current (I_{OC}):	
Device types 02 through 05 and 07 through 10	300 mA 1/
Peak collector current (I_{CP}):	
Device types 01 and 06	500 mA
Peak output current (I_{OP}):	
Device types 02 through 05 and 07 through 10	500 mA
Continuous total dissipation at 25°C ambient:	
Device types 01 and 06	1375 mW
Device types 02 through 05 and 07 through 10	1050 mW
Ambient operating temperature range	-55° to +125°C
Storage temperature range	-65° to +150°C
Junction temperature (T_J)	+175°C
Lead temperature 1/16 inch from case, (soldering, 60 seconds)	300°C

1/ Both halves of these dual circuits may conduct rated current simultaneously; however, power dissipation averaged over a short time interval must fall within the continuous power dissipation ratings.

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1.4 Recommended operating conditions.

Supply voltage range - +4.5 to +5.5 V 2/

Ambient operating temperature range- - - - - - - - - - - -55° to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θJC	Maximum θJA
14-lead dual-in-line	C	275 mW T_A = 125°C	29°C/W	91°C/W
8-lead dual-in-line	P	210 mW T_A = 125°C	26°C/W	119°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Functional schematic circuits. The functional schematic circuits shall be as specified on figure 3.

2/ For device types 01 and 06 only, the substrate (pin 8) must always be at the most negative device voltage for proper operation.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions $-55°C \leq T_A \leq 125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -12 mA	All devices	-	-1.5	V
High level input currents into A or B	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.4 V	All devices	-	40	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	All devices	-	1	mA
High level input current into G	I_{IH3}	V_{CC} = 5.5 V; V_{IN} = 2.4 V	01,05	-	80	µA
	I_{IH4}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	01,06	-	2	mA
Low level input current into A or B	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = .4 V or 5.5 V	All devices	-	-1.6	mA
Low level input current into G	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = .4 V or 5.5 V	01,06	-	-3.2	mA
High level supply current	I_{CCH1}	V_{CC} = 5.5 V; V_{IN} = 0 V	01,05	-	4	mA
	I_{CCH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	02,04,07,09	-	11	mA
	I_{CCH3}	V_{CC} = 5.5 V; V_{IN} = 0 V	05,08	-	17	mA
	I_{CCH4}	V_{CC} = 5.5 V; V_{IN} = 0 V	03	-	14	mA
	I_{CCH5}	V_{CC} = 5.5 V; V_{IN} = 0 V	10	-	12	mA
Low level supply current	I_{CCL1}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	01,06	-	11	mA
	I_{CCL2}	V_{CC} = 5.5 V; V_{IN} = 0 V	02	-	65	mA
	I_{CCL3}	V_{CC} = 5.5 V; V_{IN} = 0 V	04	-	63	mA
	I_{CCL4}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	03	-	71	mA

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \leq T_A \leq 125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Low level supply current	I_{CCL5}	V_{CC} = 5.5 V; V_{IN} = 0 V	07,09	-	76	mA
	I_{CCL5}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	08	-	76	mA
	I_{CCL7}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	05	-	79	mA
	I_{CCL8}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	10	-	85	mA
Low level output voltage	V_{OL1}	V_{CC} = 4.5 V; V_{IN} = .8 V or 4.5 V; I_{OL} = 100 mA	02,07	-	.5	V
	V_{OL2}	V_{CC} = 4.5 V; V_{IN} = 2 V; I_{OL} = 100 mA	03,08	-	.5	V
	V_{OL3}	V_{CC} = 4.5 V; V_{IN} = .8 V; I_{OL} = 100 mA	04,09	-	.5	V
	V_{OL4}	V_{CC} = 4.5 V; V_{IN} = 2 V or 0 V; I_{OL} = 100 mA	05,10	-	.5	V
	V_{OL5}	V_{CC} = 4.5 V; V_{IN} = .8 V or 4.5 V; I_{OL} = 300 mA	02,07	-	.8	V
	V_{OL6}	V_{CC} = 4.5 V; V_{IN} = 2 V; I_{OL} = 300 mA	03,08	-	.8	V
	V_{OL7}	V_{CC} = 4.5 V; V_{IN} = .8 V; I_{OL} = 300 mA	04,09	-	.8	V
	V_{OL8}	V_{CC} = 4.5 V; V_{IN} = 2 V or 0 V; I_{OL} = 300 mA	05,10	-	.8	V
Low level output voltage, TTL gate	V_{OL9}	V_{CC} = 4.5 V; V_{IN} = 2 V; I_O = 15 mA	01,06	-	.5	V
High level output current	I_{OH1}	V_{CC} = 4.5 V; V_{OH} = 30 V; V_{IN} = 2 V	02,07	-	300	µA
	I_{OH2}	V_{CC} = 4.5 V; V_{OH} = 30 V; V_{IN} = .8 or 4.5 V	03,08	-	300	µA
	I_{OH3}	V_{CC} = 4.5 V; V_{OH} = 30 V; V_{IN} = 2 V or 0 V	04,09	-	300	µA
	I_{OH4}	V_{CC} = 4.5 V; V_{OH} = 30 V; V_{IN} = .8 V	05,10	-	300	µA

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \leq T_A \leq 125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
High level output voltage, TTL gate	V_{OH}	V_{CC} = 4.5 V; V_{IN} = .8; I_{OH} = -400 µA	01,06	2.4	-	V
Short-circuit output current, TTL gate	I_{OS1}	V_{CC} = 5.5 V	01,06	-	-55	mA
	I_{OS2}	V_{CC} = 4.5 V	01,06	-18	-	mA
Collector-base breakdown voltage	V_{CBO1}	I_C = 100 µA; I_E = 0	01	35	-	V
	V_{CBO2}	I_C = 100 µA; I_E = 0	06	40	-	V
Collector-emitter breakdown voltage	V_{CER1}	I_C = 100 µA; R_{BE} = 500 Ω	01	35	-	V
	V_{CER2}	I_C = 100 µA; R_{BE} = 500 Ω	06	40	-	V
Collector-emitter breakdown voltage	V_{CEO1}	I_C = 10 mA	06	25	-	V
Emitter-base breakdown voltage	V_{EBO}	I_E = 100 µA; I_C = 0	01,06	5	-	V
Static forward current transfer ratio	h_{FE1}	V_{CE} = 3 V; I_C = 100 mA; T_A = 25°C, 125°C, V_S = 4 V	01,06	25	-	-
	h_{FE2}	V_{CE} = 3 V; I_C = 300 mA; T_A = 25°C, 125°C, V_S = 6 V	01,06	30	-	-
	h_{FE3}	V_{CE} = 3 V; I_C = 100 mA; T_A = -55°C, V_S = 4 V	01,06	10	-	-
	h_{FE4}	V_{CE} = 3 V; I_C = 300 mA; T_A = -55°C, V_S = 5 V	01,06	15	-	-
Base-emitter voltage	V_{BE1}	I_B = 10 mA; I_C = 100 mA	01,06	-	1.2	V
	V_{BE2}	I_B = 30 mA; I_C = 300 mA	01,06	-	1.4	V
Collector-emitter saturation voltage	V_{CESAT1}	I_B = 10 mA; I_C = 100 mA	01,06	-	.5	V
	V_{CESAT2}	I_B = 30 mA; I_C = 300 mA	01,06	-	.9	V

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55^{\circ}C \leq T_A \leq 125^{\circ}C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Delay time	t_D	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{BE} = -1 V; T_A = 25°C I_B = 20 mA $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01,06	- -	15 22.5	ns "
Rise time	t_R	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{BE} = -1 V; T_A = 25°C I_B = 20 mA $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01,06	- -	20 30	" "
Storage time	t_S	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{BE} = -1 V; T_A = 25°C I_B = 20 mA $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01	- -	15 22.5	" "
	t_S	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{BE} = -1 V; T_A = 25°C I_B = 20 mA $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	06	- -	23 34.5	" "
Fall time	t_F	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{BE} = -1 V; T_A = 25°C I_B = 20 mA $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01,06	- -	15 22.5	" "
Propagation delay time (low to high level) output) TTL gate	t_{PLH1}	C_L = 100 pF; R_L = 400 Ω; V_{CC} = 4.5 V; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01,06	- -	30 45	" "
Propagation delay time (low to high level output)	t_{PLH2}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{CC} = 4.5 V; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01	- -	30 45	" "
	t_{PLH3}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; V_{CC} = 4.5 V; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	06	- -	65 90	" "
Propagation delay time (high to low level output) TTL gate	t_{PHL1}	C_L = 100 pF; R_L = 400 Ω; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01,06	- -	15 22.5	" "
Propagation delay time (high to low level output)	t_{PHL2}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01	- -	30 45	" "
	t_{PHL3}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01	- -	50 75	" "

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55^{\circ}C \leq T_A \leq 125^{\circ}C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Transition time (low to high level output)	t_{TLH1}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01	- -	15 22.5	" "
	t_{TLH2}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	06	- -	20 30	" "
Transition time (high to low level output)	t_{THL1}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	01	- -	15 22.5	ns "
	t_{THL2}	I_C = 200 mA; R_L = 50 Ω; C_L = 100 pF; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	06	- -	20 30	" "
Propagation delay time (low to high level output)	t_{PLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	02,04	- -	30 45	" "
	t_{PLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	03,05	- -	35 55	" "
	t_{PLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	07,09	- -	55 65	" "
	t_{PLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	08,10	- -	65 90	" "
Propagation delay (high to low level output)	t_{PHL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	02,04	- -	30 45	" "
	t_{PHL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	03,05	- -	35 55	" "
	t_{PHL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	07,09	- -	40 60	" "
	t_{PHL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55^{\circ}C \leq T_A \leq 125^{\circ}C$	08,10	- -	50 75	" "

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $-55°C \le T_A \le 125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Transition time (low to high level output)	t_{TLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55°C \le T_A \le 125°C$	02,03,04,05	- -	14 18.5	ns ns
	t_{TLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55°C \le T_A \le 125°C$	07,10	- -	20 26.5	ns ns
	t_{TLH}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55°C \le T_A \le 125°C$	08,09	- -	25 33.5	ns ns
Transition time (high to low level output)	t_{THL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55°C \le T_A \le 125°C$	02,03,04,05	- -	20 25	ns ns
	t_{THL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C $-55°C \le T_A \le 125°C$	07,08,10	- -	20 25	ns ns
	t_{THL}	V_{CC} = 4.5 V; C_L = 100 pF; R_L = 50 Ω; I_O = 200 mA; T_A = 25°C	09	-	25	ns

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3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.2.6 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics of table I apply over the full recommended ambient temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 53 (see MIL-M-38510, Appendix E).

TABLE II. Electrical test requirements.

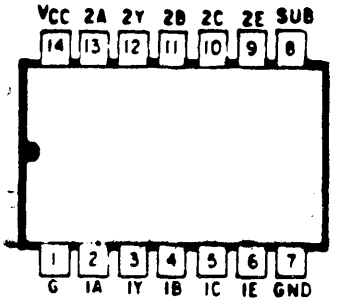
MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,9, 10,11	1,2,3,9
Group C end-point and group B class S, electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1

(*) indicates PDA applies to subgroup 1 (see 4.2c).

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Device Types 01, 06

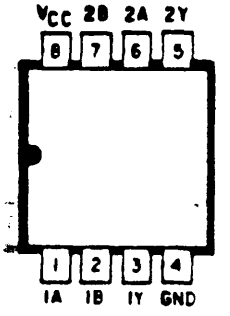
(Case C)



14 LEAD DIP
(TOP VIEW)

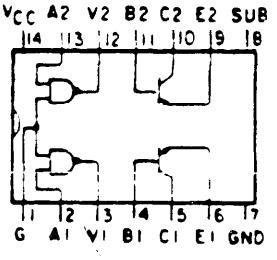
Device Types 02-05, 07-10

(Case P)

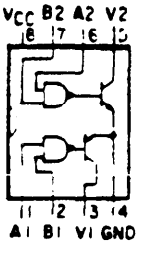


8 LEAD DIP
(TOP VIEW)

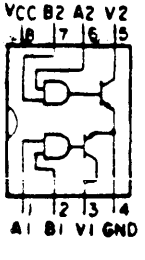
Device Type 01, 06



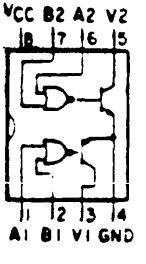
Device Type 02, 07



Device Type 03, 08



Device Type 04, 09



Device Type 05, 10

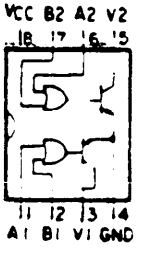


FIGURE 1. Logic diagrams and terminal connections.

Device type 01, 06

Truth Table

A	G	Y
0	0	1
0	1	1
1	0	1
1	1	0

Device type 02, 07

Truth Table

A	B	Y	OUTPUT
0	0	0	ON
0	1	0	ON
1	0	0	ON
1	1	1	OFF

Device type 03, 08

Truth Table

A	B	Y	OUTPUT
0	0	1	OFF
0	1	1	OFF
1	0	1	OFF
1	1	0	ON

Device type 04, 09

Truth Table

A	B	Y	OUTPUT
0	0	0	ON
0	1	1	OFF
1	0	1	OFF
1	1	1	OFF

Device type 05, 10

Truth Table

A	B	Y	OUTPUT
0	0	1	OFF
0	1	0	ON
1	0	0	ON
1	1	0	ON

FIGURE 2. Truth tables.

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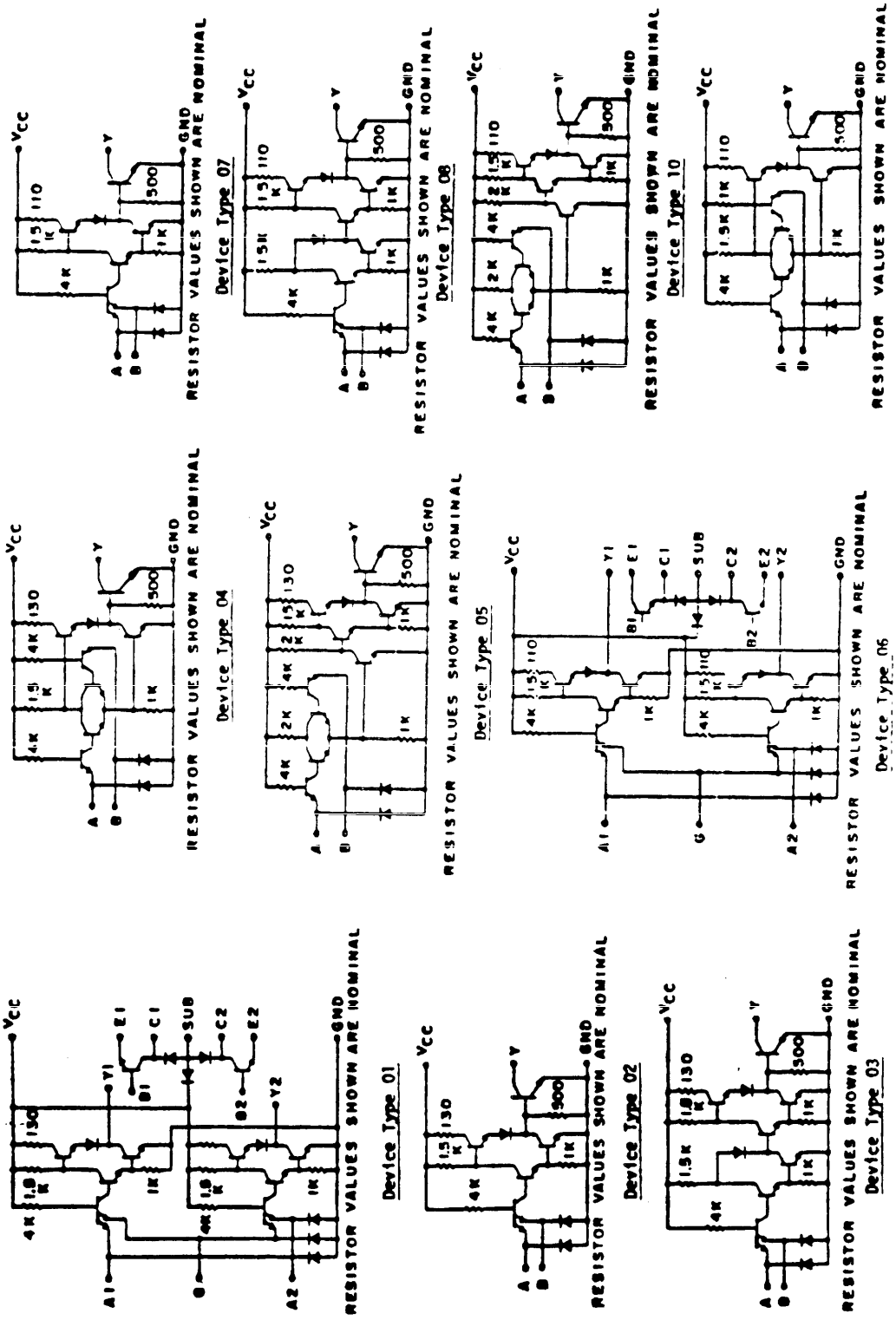


FIGURE 3. Functional schematic circuits.

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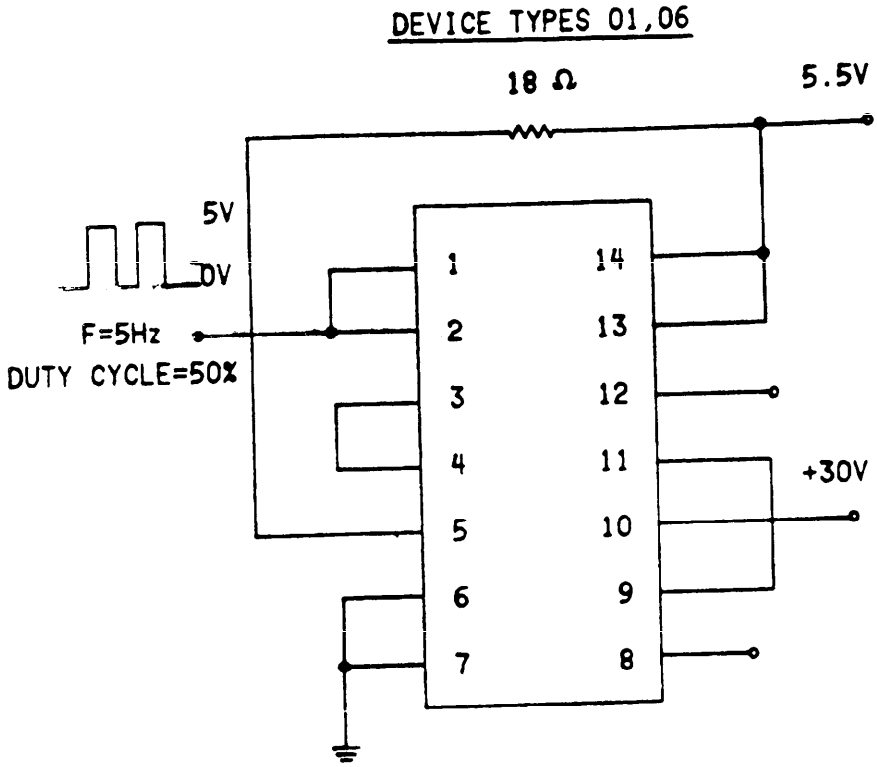


FIGURE 4. Test circuit, burn-in and operating life test.

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VERTICAL PINTLE LOAD: 600 LBS

DEVICE TYPES 02,03,04,05,07,08,09,10

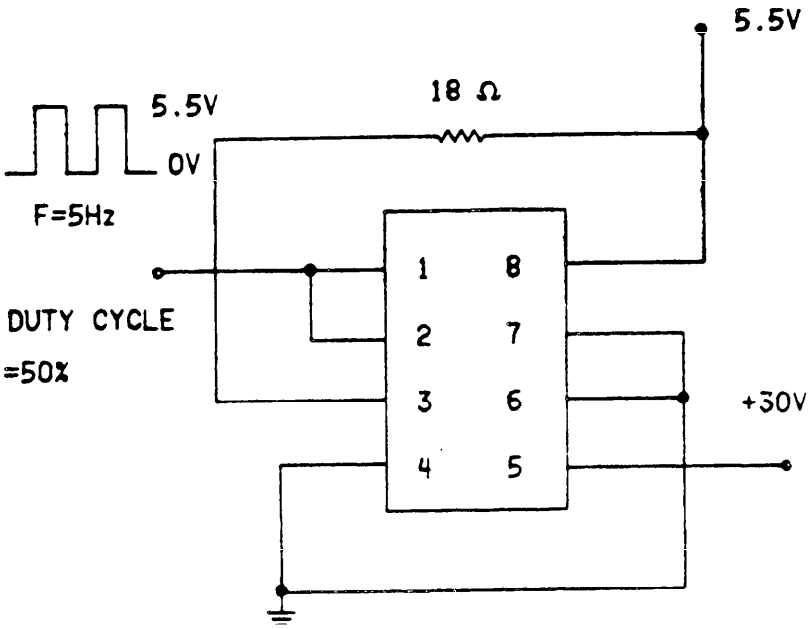


FIGURE 4. Test circuit, burn-in and operating life test - Continued.

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DEVICE TYPES 01,06

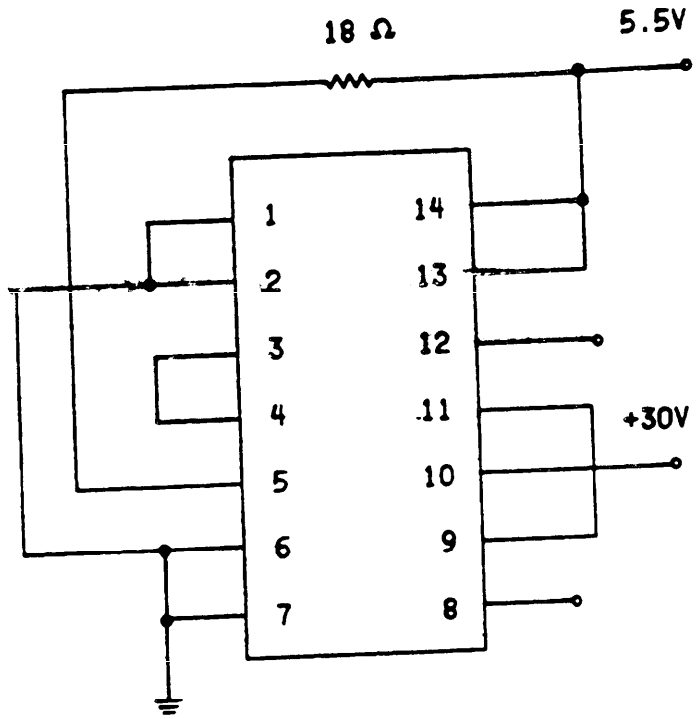


FIGURE 5. Test circuit, burn-in (steady state power and reverse bias) and operating life test.

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DEVICE TYPES 02,03,04,05,07,08,09,10

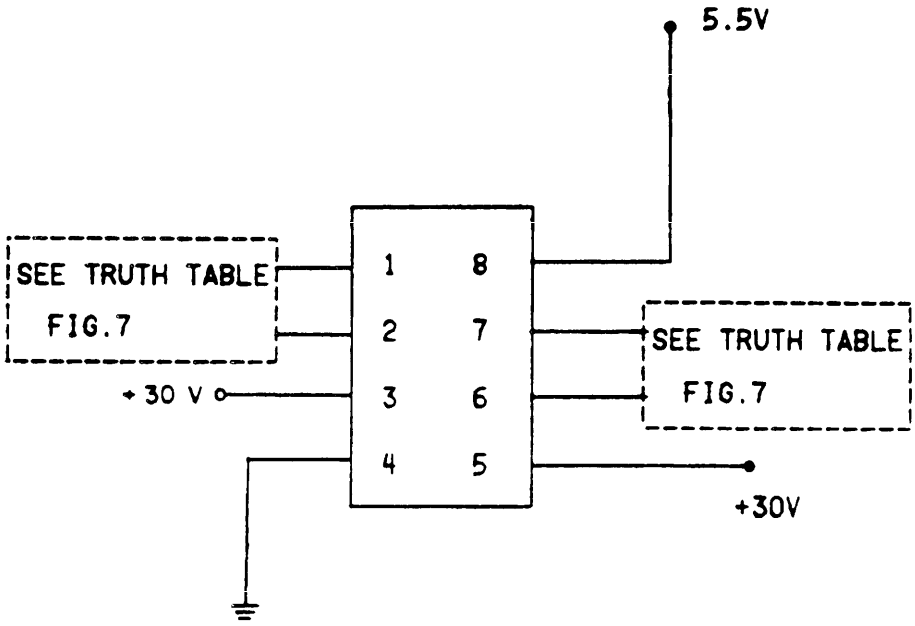


FIGURE 5. Test circuit, burn-in (steady state power and reverse bias) and operating life test - Continued.

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this appendix

This section is not applicable to

DEVICE TYPES 01,06

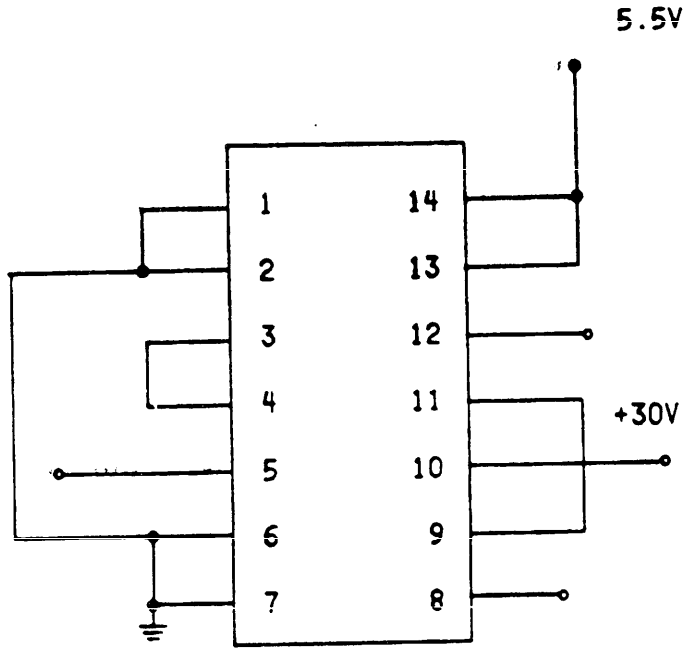


FIGURE 6. Accelerated burn-in and operating life test circuit.

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DEVICE TYPES 02,03,04,05,07,08,09,10

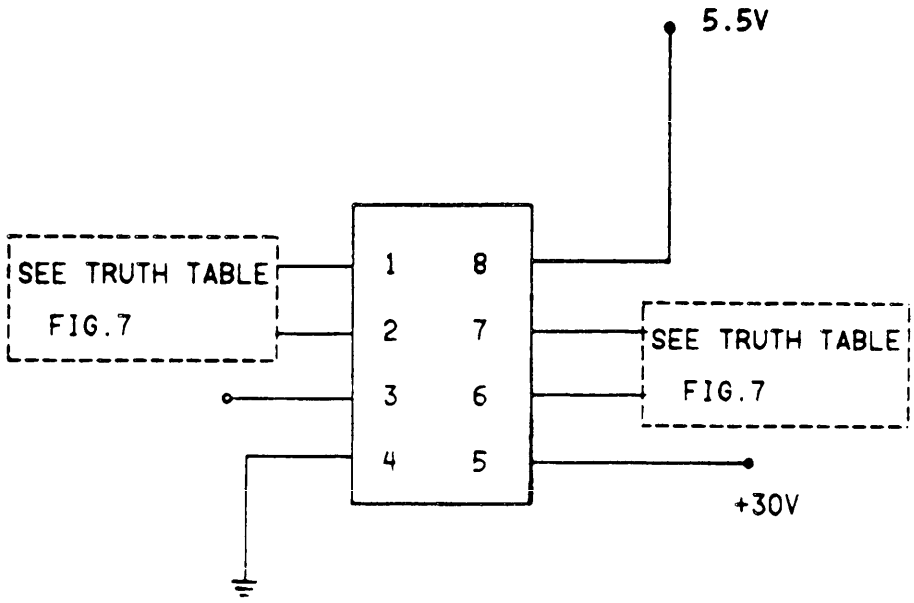


FIGURE 6. Accelerated burn-in and operating life test circuit - Continued.

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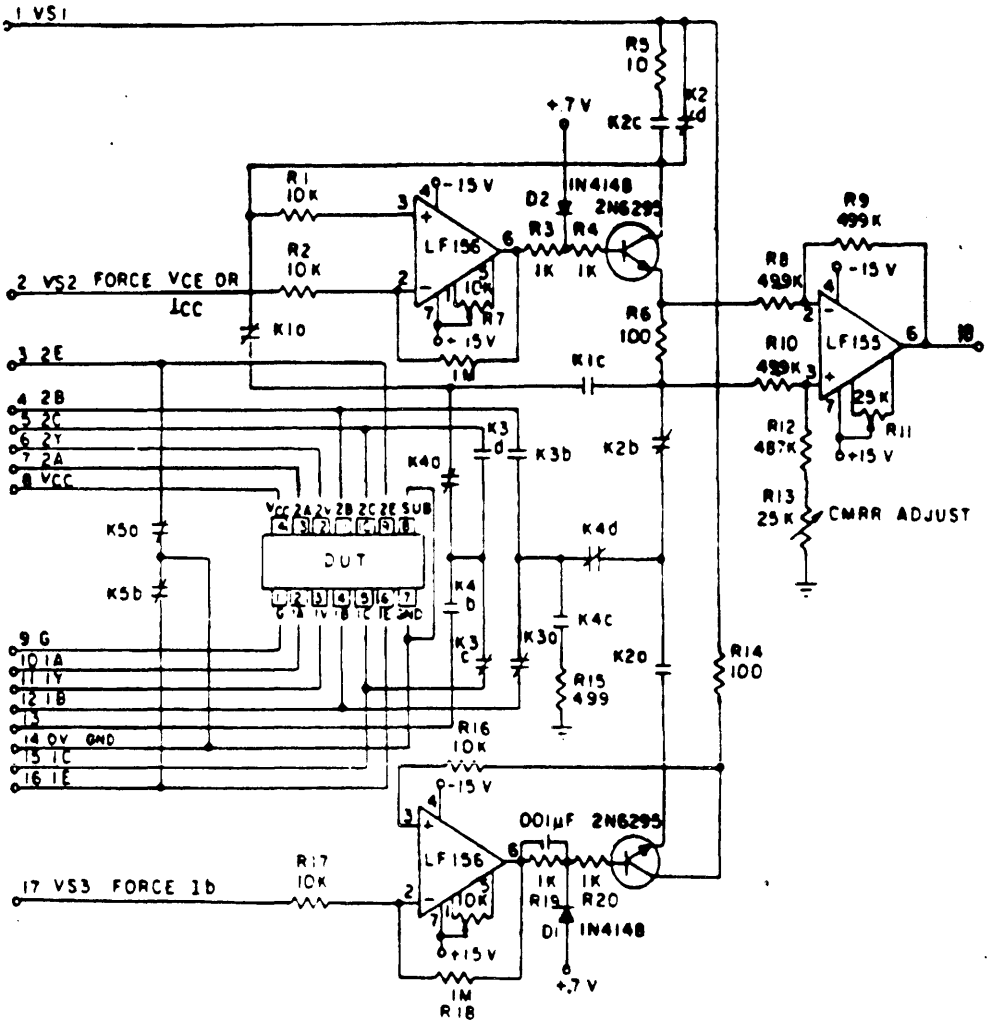
DEVICE TYPES 02,04,07,09						DEVICE TYPES 03,05,08,10					
INPUTS				OUTPUTS		INPUTS				OUTPUTS	
1(A1)	2(A2)	6(B1)	7(B2)	A	B	1(A1)	2(A2)	6(B1)	7(B2)	A	B
L	L	H	H	L	H	H	H	L	L	L	H
L = 0 V H = 5 V						L = 0 V H = 5 V					

FIGURE 7. Truth table for figure 6 burn-in test circuits.

DEVICE TYPES 02,04,07,09						DEVICE TYPES 03,05,08,10					
INPUTS				OUTPUTS		INPUTS				OUTPUTS	
1(A1)	2(A2)	6(B1)	7(B2)	A	B	1(A1)	2(A2)	6(B1)	7(B2)	A	B
H	H	H	H	H	H	L	L	L	L	H	H
L = 0 V H = 5 V						L = 0 V H = 5 V					

FIGURE 7. Truth table for figure 5 burn-in test circuits.

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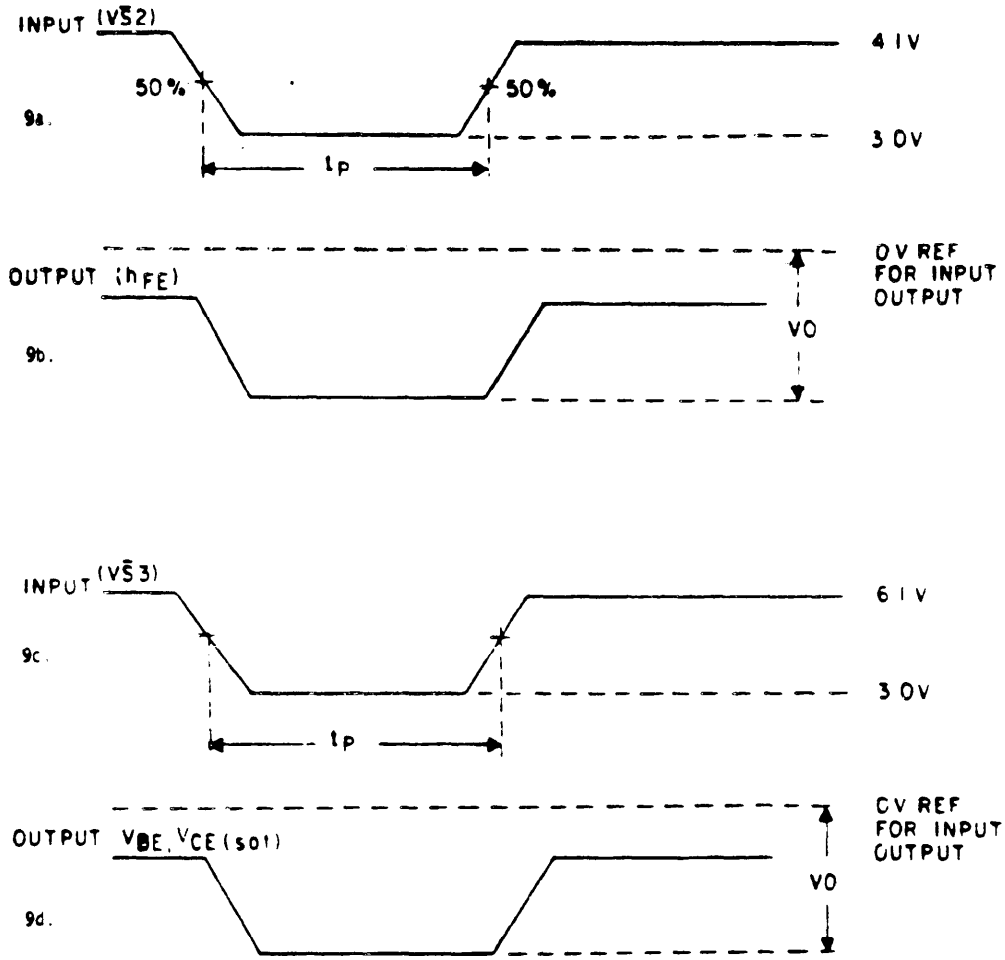


NOTES:

1. The output offset voltage shall be adjusted to zero volts with the DUT removed. The operational amplifier stabilization networks may vary with test adapter construction. Alternate drive circuits for the 2N6295 may be used to develop the proper forcing currents and input voltage pulses. These circuits shall require the approval of the qualifying activity.
2. Relay switch positions are defined in table III.
3. Resistors R5, R6, R14 shall have a tolerance ≤0.1% for device types 01, 06.
4. Reference figure 9 for input and output waveforms for device types 01, 06.

FIGURE 8. Test circuit for static tests, device types 01 and 06.

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NOTES: 1. The pulse generator shall have the following characteristics:

PRR = 1 KHz, t_p = 20 µs, Zout = 50Ω. t_r, t_f = 10 ns.

2. All V_O measurements are referenced to 0 V GND.
3. Use figures 9a and 9b for tests 23-26 and figures 9a, 9c, and 9d for tests 27-34.

FIGURE 9. h_{FE}, V_{BE}, $V_{CE(sat)}$ waveforms for table III, device types 01 and 06.

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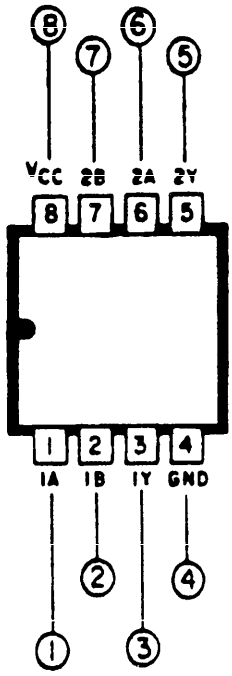
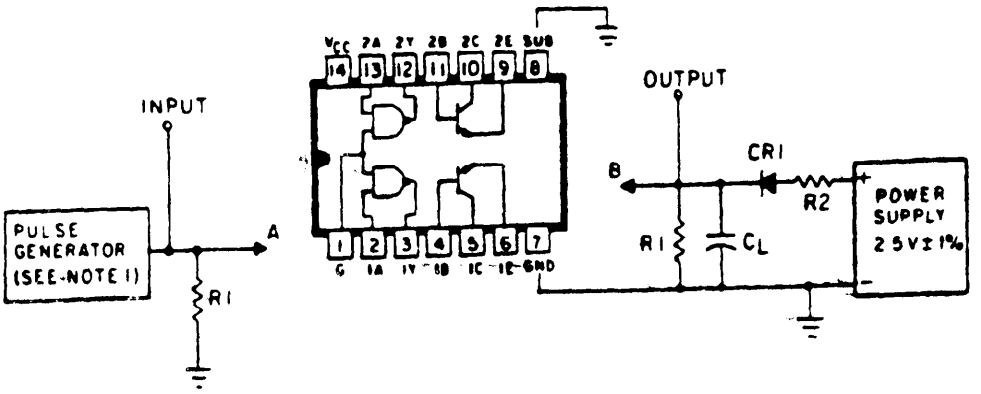
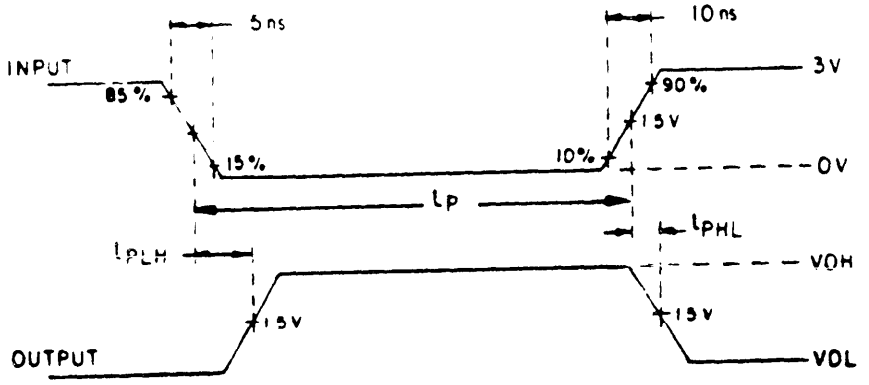


FIGURE 10. Test circuit for static tests, device types 02 through 05 and 07 through 10.

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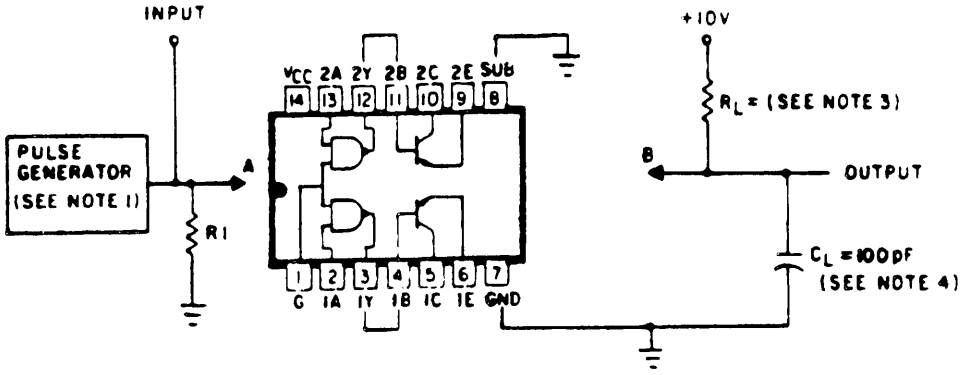
NOTES: 1. The pulse generator shall have the following characteristics:

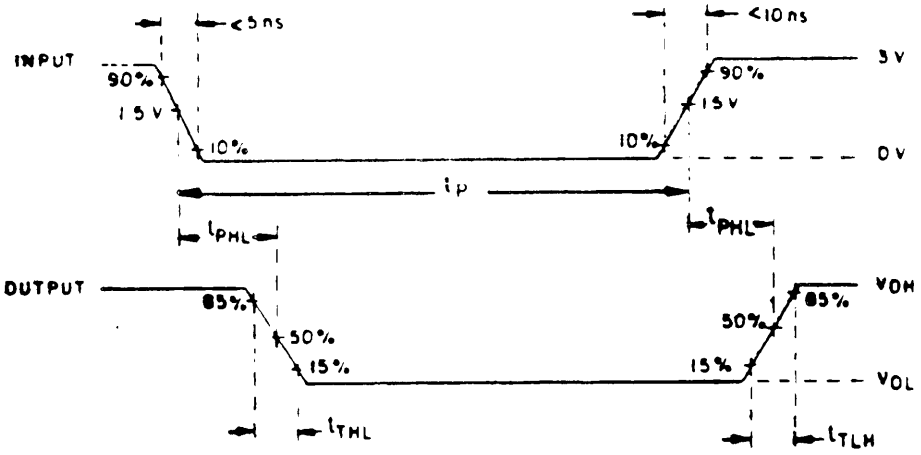
PRR = 1 MHz, t_p = 0.5 µs, $Z_{out} \cong 50\Omega$.

2. R1 = 6.04 kΩ ±1 percent, 1/8 watt.
3. C_L = 100 pF, ±5 pF (including probe and parasitic capacitance).
4. Select R2 for a current flow of 16 mA ±1% out of Point B with Point L held at 0.5 volts.
5. CR1 = 1N4150 or equivalent.
6. R1 = 51Ω ±5% carbon.

FIGURE 11. Propagation delay time waveforms (TTL gates only), for device types 01 and 06

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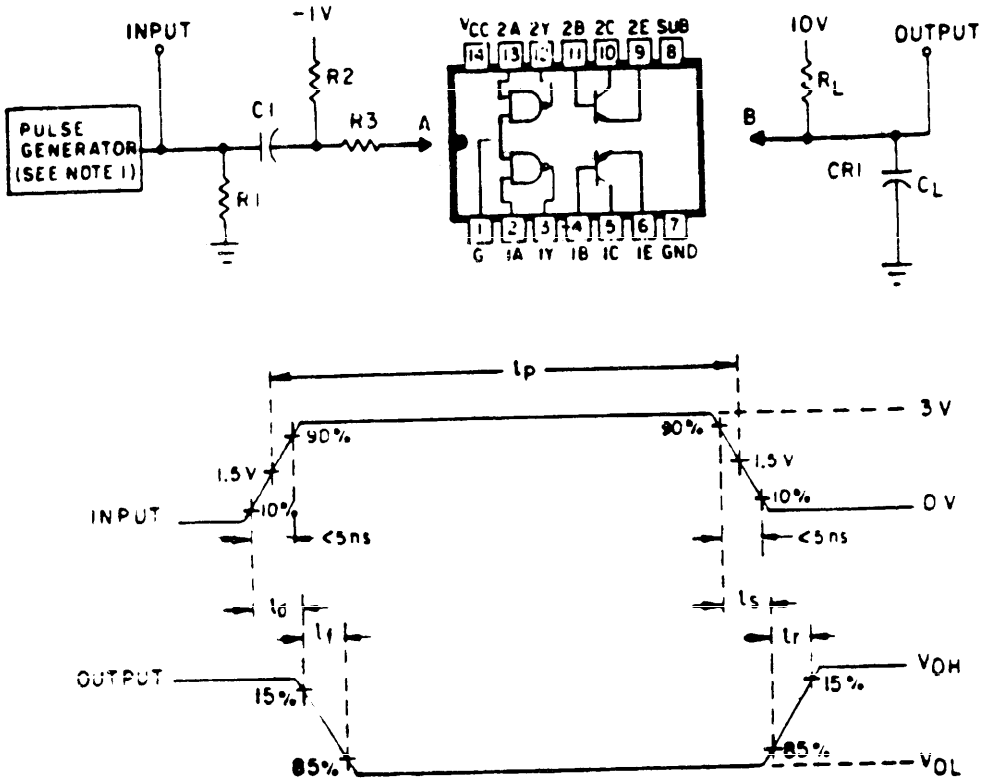


NOTES:
1. The pulse generator shall have the following characteristics:

   PRR = 1 MHz, t_p = 0.5 µs, Z_{out} ≅ 50Ω.
2. When testing device 01 or 06, connect output Y to transistor base and ground the substrate terminal.
3. R_L = 47Ω±5% carbon.
4. C_L = 100 pF minimum, including probe and jig capacitance.
5. R1 = 51Ω ±5% carbon.

FIGURE 12. Switching time waveforms (TTL gates and transistors combined) for device types 01 and 06.

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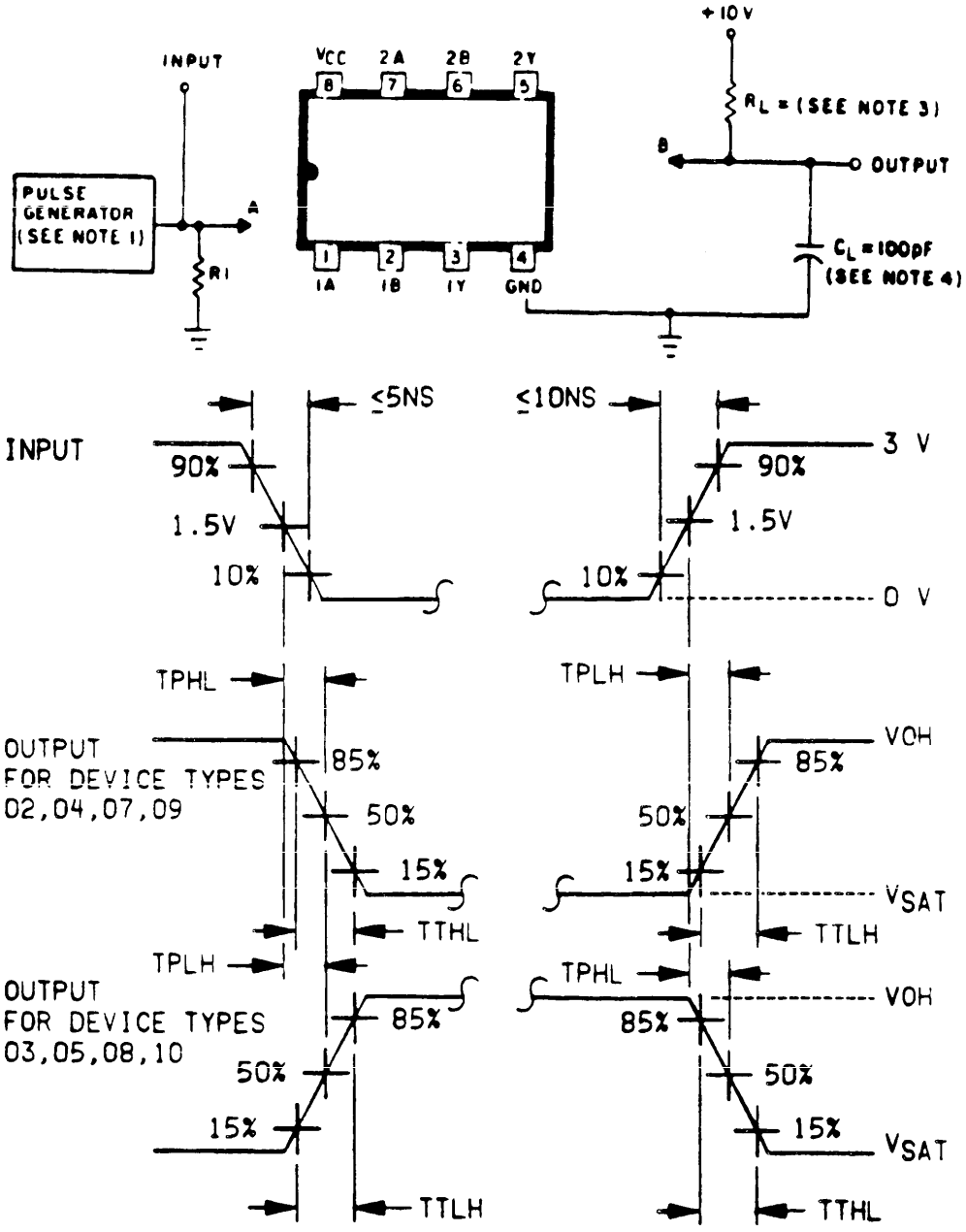


NOTES:
1. The pulse generator shall have the following characteristics:

   PRR = 1 MHz, t_p = 0.3 µs, Z_{out} ≅ 50Ω.
2. C_L = 100 pF minimum, including probe and jig capacitance.
3. R1 = 62Ω ±5% carbon.
4. R2 = 1 KΩ ±5% carbon.
5. C1 = 0.1 µF ±5%.
6. R_L = 47Ω ±5% carbon.
7. R3 = 51Ω ±5% carbon.
8. All voltages have a tolerance of ±1% of nominal.

FIGURE 13. Switching times for device types 01 and 06. (transistors only).

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NOTES:
1. Pulse generator shall have the following characteristics: PRR ≤ 1 MHz, Z_{OUT} = 50 , t_r ≤ 10 ns, t_f ≤ 5 ns.
2. R1 = 51Ω ±5%.
3. R_L = 47Ω ±5%.
4. C_L = 100 pF ±10%, including probe and jig capacitance.

FIGURE 14. Switching time waveforms for device types 02 through 05 and 07 through 10.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	1	2	17	12	20	16	4	3	5	8	9	10	11	7	6	Relays energized	Measured terminal			Equations	Note	Test limits		Unit
																			No.	Value	Unit			Min	Max	
1 T_A = 25°C	V_{IC}	1										4.5 V	-12 mA	4.5 V		4.5 V		None	9	E1	V	V_{IC} = E1	See figure 8 for test circuit	-	-1.5	V
	V_{IC}	2										4.5 V	4.5 V	-12 mA		4.5 V		"	10	E2	V	V_{IC} = E2		-	-1.5	V
	V_{IC}	3										4.5 V	4.5 V	4.5 V		-12 mA		"	7	E3	V	V_{IC} = E3	"	-	-1.5	V
	I_{IH1}	4										5.5 V	GND	2.4 V		GND		"	10	I1	A	I_{IH1} = I1	"	-	40	µA
	I_{IH1}	5										"	"	GND		2.4 V		"	7	I2	"	I_{IH1} = I2	"	-	40	µA
	I_{IH2}	6										"	"	5.5 V		GND		"	10	I3	"	I_{IH2} = I3	"	-	1	mA
	I_{IH2}	7										"	"	GND		5.5 V		"	7	I4	"	I_{IH2} = I4	"	-	1	mA
	I_{IH3}	8										"	2.4 V	GND		GND		"	9	I5	"	I_{IH3} = I5	"	-	80	µA
	I_{IH4}	9										"	5.5 V	GND		GND		"	9	I6	"	I_{IH4} = I6	"	-	2	mA
	I_{IL1}	10										"	5.5 V	0.4 V		5.5 V		"	10	I7	"	I_{IL1} = I7	"	-	-1.6	"
	I_{IL1}	11										"	5.5 V	5.5 V		0.4 V		"	7	I8	"	I_{IL1} = I8	"	-	-1.6	"
	I_{IL2}	12										"	0.4 V	5.5 V		5.5 V		"	9	I9	"	I_{IL2} = I9	"	-	-3.2	"
	I_{CCH1}	13										"	GND	GND		GND		"	8	I10	"	I_{CCH1} = I10	"	-	4	"
	I_{CCL1}	14										"	5.5 V	5.5 V		5.5 V		"	8	I11	"	I_{CCL1} = I11	"	-	11	"
	V_{OL9}	15										4.5 V	2 V	2 V	I_{OL} = 16 mA			"	11	E4	V	V_{OL9} = E4	"	-	0.5	V
	V_{OL9}	16										"	2 V			2 V	I_{OL} = 16 mA	"	6	E5	"	V_{OL9} = E5	"	-	0.5	"
	V_{OH}	17										"	4.5 V	0.8 V	I_{OH} = -800 µA			"	11	E6	"	V_{OH} = E6	"	2.4	-	"
	V_{OH}	18										"	4.5 V			0.8 V	I_{OH} = -800 µA	"	6	E7	"	V_{OH} = E7	"	2.4	-	"
	I_{OS1}	19										5.5 V	GND	GND	GND			"	11	I12	A	I_{OS1} = I12	"	-	-55	mA
	I_{OS1}	20										5.5 V	"			GND	GND	"	8	I13	"	I_{OS1} = I13	"	-	-55	"
	I_{OS2}	21										4.5 V	"	GND	GND			"	11	I14	"	I_{OS2} = I14	"	-18	-	"
	I_{OS2}	22										4.5 V	"			GND	GND	"	6	I15	"	I_{OS2} = I15	"	-18	-	"
	h_{FE1}	23	4 V	$\overline{V_{S2}}$														"	10 to 14	E8	V	h_{FE1} = $\frac{E8}{1000}$	See figures 8 and 9 for test circuit and waveforms	25	-	-
	h_{FE1}	24	4 V	"														3	"	E9	"	h_{FE1} = $\frac{E9}{1000}$		25	-	-
	h_{FE2}	25	6 V	"														None	"	E10	"	h_{FE2} = $\frac{E10}{3000}$	"	30	-	-
	h_{FE2}	26	6 V	"														3	"	E11	"	h_{FE2} = $\frac{E11}{3000}$	"	30	-	-
	V_{BE1}	27	4 V	"	$\overline{V_{S3}}$													1,2	12 to 16	E12	"	V_{BE1} = E12	"	-	1.2	V
	V_{BE1}	28	4 V	"	$\overline{V_{S3}}$													1,2,3	11 to 10	E13	"	V_{BE1} = E13	"	-	1.2	V

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	15	16	4	3	5	8	9	10	11	7	6	Relays energized	Measured terminal No.	Measured terminal Value	Measured terminal Unit	Equations	Note	Test limits Min	Test limits Max	Unit
			V_{S1}	V_{S2}	V_{S3}	2B	2C	2E	2H	2K	2L	V_{CC}	G	1A	1Y	2A	2Y									
T_A = 25°C 1	V_{BE2}	29	6 V	V_{S2}	V_{S3}													1,2	12 to 16	E14	V	V_{BE2} = E14	See figures 8 and 9 for test circuit and waveforms	-	1.4	V
	V_{BE2}	30	6 V	"	"													1,2,3	11 to 10	E15	"	V_{BE2} = E15	"	-	1.4	"
	V_{CESAT1}	31	4 V	"	"													1,2	5 to 6	E16	"	V_{CESAT1} = E16	"	-	0.5	"
	V_{CESAT}	32	4 V	"	"													1,2,3	10 to 9	E17	"	V_{CESAT1} = E17	"	-	0.5	"
	V_{CESAT2}	33	6 V	"	"													1,2	5 to 6	E18	"	V_{CESAT2} = E18	"	-	0.8	"
	V_{CESAT2}	34	6 V	"	"													1,2,3	10 to 9	E19	"	V_{CESAT2} = E19	"	-	0.8	"
	V_{CBO1}	35				GND	100 µA											3	15 to 12	E20	"	V_{CBO1} = E20	See figure 8 for test circuit	35	-	"
	V_{CBO1}	36							GND		100 µA							4	5 to 3	E21	"	V_{CBO1} = E21		35	-	"
	V_{CER1}	37					100 µA	GND										4	15 to 16	E22	"	V_{CER1} = E22	"	35	-	"
	V_{CER1}	38								GND	100 µA							3,4	5 to 3	E23	"	V_{CER1} = E23	"	35	-	"
	V_{EBO}	39				GND		100 µA										3,4	16 to 12	E24	"	V_{EBO} = E24	"	5	-	"
	V_{EBO}	40							GND	100 µA								4	3 to 4	E25	"	V_{EBO} = E25	"	5	-	"
T_A = 125°C 2	V_{IC}	41										4.5 V	-12 mA	4.5 V		4.5 V		None	9	E1	"	V_{IC} = E1	"	-	-1.5	"
	V_{IC}	42										4.5 V	4.5 V	-12 mA		4.5 V		"	10	E2	"	V_{IC} = E2	"	-	-1.5	"
	V_{IC}	43										4.5 V	4.5 V	4.5 V		-12 mA		"	7	E3	"	V_{IC} = E3	"	-	-1.5	"
	I_{IH1}	44										5.5 V	GND	2.4 V		GND		"	10	I1	A	I_{IH1} = I1	"	-	80	µA
	I_{IH1}	45										"	"	GND		2.4 V		"	7	I2	"	I_{IH1} = I2	"	-	80	µA
	I_{IH2}	46										"	"	5.5 V		GND		"	10	I3	"	I_{IH2} = I3	"	-	1	mA
	I_{IH2}	47										"	"	GND		5.5 V		"	7	I4	"	I_{IH2} = I4	"	-	1	mA
	I_{IH3}	48										"	2.4 V	GND		GND		"	9	I5	"	I_{IH3} = I5	"	-	80	µA
	I_{IH4}	49										"	5.5 V	GND		GND		"	9	I6	"	I_{IH4} = I6	"	-	2	mA
	I_{IL1}	50										"	5.5 V	0.4 V		5.5 V		"	10	I7	"	I_{IL1} = I7	"	-	-1.6	"
	I_{IL1}	51										"	5.5 V	5.5 V		0.4 V		"	7	I8	"	I_{IL1} = I8	"	-	-1.6	"
	I_{IL2}	52										"	0.4 V	5.5 V		5.5 V		"	9	I9	"	I_{IL2} = I9	"	-	-3.2	"
	I_{CCH1}	53										"	GND	GND		GND		"	8	I10	"	I_{CCH1} = I10	"	-	4	"
	I_{CCL1}	54										"	5.5 V	5.5 V		5.5 V		"	8	I11	"	I_{CCL1} = I11	"	-	11	"
	V_{OL9}	55										4.5 V	2 V	2 V	I_{OL} = 16 mA			"	11	E4	V	V_{OL9} = E4	"	-	0.5	V
	V_{OL9}	56										"	2 V			2 V	I_{OL} = 16 mA	"	6	E5	"	V_{OL9} = E5	"	-	0.5	"
	V_{OH}	57										"	4.5 V	0.8 V	I_{OH} = -400 µA			"	11	E6	"	V_{OH} = E6	"	2.4	-	"
	V_{OH}	58										"	4.5 V			0.8 V	I_{OH} = -400 µA	"	6	E7	"	V_{OH} = E7	"	2.4	-	"

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	15	16	8	3	5	8	9	10	11	7	6	Relays energized	Measured terminal			Equations	Note	Test limits		Unit
			V_{S1}	V_{S2}	V_{S3}	1B	1C	1E	2B	2E	2C	V_{CC}	G	1A	1Y	2A	2Y		No.	Voltage	Units			Min	Max	
2 T_A = 125°C	I_{OS1}	59										5.5 V	GND	GND	GND			None	11	I12	A	I_{OS1} = I12	See figure 8 for test circuit	-	-55	mA
	I_{OS1}	60										5.5 V	"			GND	GND	"	6	I13	"	I_{OS1} = I13		-	-55	"
	I_{OS2}	61										4.5 V	"	GND	GND			"	11	I14	"	I_{OS2} = I14	"	-18	-	"
	I_{OS2}	62										4.5 V	"			GND	GND	"	6	I15	"	I_{OS2} = I15	"	-18	-	"
	h_{FE1}	63	4 V	$\overline{V_{S2}}$									"					"	18 to 14	E8	V	$h_{FE1} = \frac{E8}{1000}$	See figures 8 and 9 for test circuit and waveforms	25	-	-
	h_{FE1}	64	4 V	"														3	"	E9	"	$h_{FE1} = \frac{E9}{1000}$	"	25	-	-
	h_{FE2}	65	6 V	"														None	"	E10	"	$h_{FE2} = \frac{E10}{3000}$	"	30	-	-
	h_{FE2}	66	6 V	"														3	"	E11	"	$h_{FE2} = \frac{E11}{3000}$	"	30	-	-
	V_{BE1}	67	4 V	"	$\overline{V_{S3}}$													1,2	12 to 16	E12	"	V_{BE1} = E12	"	-	1.2	V
	V_{BE1}	68	4 V	"	"													1,2,3	11 to 10	E13	"	V_{BE1} = E13	"	-	1.2	"
	V_{BE2}	69	6 V	"	"													1,2	12 to 16	E14	"	V_{BE2} = E14	"	-	1.4	"
	V_{BE2}	70	6 V	"	"													1,2,3	11 to 10	E15	"	V_{BE2} = E15	"	-	1.4	"
	V_{CESAT1}	71	4 V	"	"													1,2	5 to 6	E16	"	V_{CESAT1} = E16	"	-	0.5	"
	V_{CESAT1}	72	4 V	"	"													1,2,3	10 to 9	E17	"	V_{CESAT1} = E17	"	-	0.5	"
	V_{CESAT2}	73	6 V	"	"													1,2	5 to 6	E18	"	V_{CESAT2} = E18	"	-	0.8	"
	V_{CESAT2}	74	6 V	"	"													1,2,3	10 to 9	E19	"	V_{CESAT2} = E19	"	-	0.8	"
	V_{CBO1}	75				GND	100 µA											3	15 to 12	E20	"	V_{CBO1} = E20	See figure 8 for test circuit	35	-	"
	V_{CBO1}	76							GND		100 µA							4	5 to 3	E21	"	V_{CBO1} = E21		35	-	"
	V_{CER1}	77					100 µA	GND										4	15 to 16	E22	"	V_{CER1} = E22	"	35	-	"
	V_{CER1}	78								GND	100 µA							3,4	5 to 3	E23	"	V_{CER1} = E23	"	35	-	"
	V_{EBO}	79				GND		100 µA										3,4	16 to 12	E24	"	V_{EBO} = E24	"	5	-	"
	V_{EBO}	80							GND	100 µA								4	3 to 4	E25	"	V_{EBO} = E25	"	5	-	"
3 T_A = -55°C	V_{IC}	81										4.5 V	-12 mA	4.5 V		-.5 V		None	9	E1	"	V_{IC} = E1	"	-	-1.5	"
	V_{IC}	82										4.5 V	4.5 V	-12 mA		4.5 V			10	E2	"	V_{IC} = E2	"	-	-1.5	"
	V_{IC}	83										4.5 V	4.5 V	4.5 V		-12 mA			7	E3	"	V_{IC} = E3	"	-	-1.5	"
	I_{IH1}	84										5.5 V	GND	2.4 V		GND			10	I1	A	I_{IH1} = I1	"	-	40	µA
	I_{IH1}	85										"	"	GND		2.4 V			7	I2	"	I_{IH1} = I2	"	-	40	µA
	I_{IH2}	86										"	"	5.5 V		GND			10	I3	"	I_{IH2} = I3	"	-	1	mA
	I_{IH2}	87										"	"	GND		5.5 V			7	I4	"	I_{IH2} = I4	"	-	1	mA
	I_{IH3}	88										"	2.4 V	GND		GND			9	I5	"	I_{IH3} = I5	"	-	80	µA
	I_{IH4}	89										"	5.5 V	GND		GND			9	I6	"	I_{IH4} = I6	"	-	2	mA

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	18	16	4	3	5	8	9	10	11	7	6	Relays ener-gized	Measured constraint			Equations	Note	Test limits		Unit
																			No.	Value	Unit			Min	Max	
T_A = -55°C	I_{IL1}	90										5.5 V	5.5 V	0.4 V		5.5 V		None	10	I7	A	I_{IL1} - I7	See figure 8 for test circuit	-	-1.6	mA
	I_{IL1}	91										"	5.5 V	5.5 V		0.4 V		"	7	I8	"	I_{IL1} - I8	"	-	-1.6	"
	I_{IL2}	92										"	0.4 V	5.5 V		5.5 V		"	9	I9	"	I_{IL2} - I9	"	-	-3.2	"
	I_{CCH1}	93										"	GND	GND		GND		"	8	I10	"	I_{CCH1} - I10	"	-	4	"
	I_{CCL1}	94										"	5.5 V	5.5 V		5.5 V		"	8	I11	"	I_{CCL1} - I11	"	-	11	"
	V_{OL9}	95										4.5 V	2 V	2 V	I_{OL} = 16 mA			"	11	E4	V	V_{OL9} - E4	"	-	0.5	V
	V_{OL9}	96										"	2 V			2 V	I_{OL} = 16 mA	"	6	E5	"	V_{OL9} - E5	"	-	0.5	"
	V_{OH}	97										"	4.5 V	0.8 V	I_{OH} = -400 µA			"	11	E6	"	V_{OH} - E6	"	2.4	-	"
	V_{OH}	98										"	4.5 V			0.8 V	I_{OH} = -400 µA	"	6	E7	"	V_{OH} - E7	"	2.4	-	"
	I_{OS1}	99										5.5 V	GND	GND	GND			"	11	I12	A	I_{OS1} - I12	"	-	-55	mA
	I_{OS1}	100										5.5 V	"			GND	GND	"	6	I13	"	I_{OS1} - I13	"	-	-55	"
	I_{OS2}	101										4.5 V	"	GND	GND			"	11	I14	"	I_{OS2} - I14	"	-18	-	"
	I_{OS2}	102										4.5 V	"			GND	GND	"	6	I15	"	I_{OS2} - I15	"	-18	-	"
	h_{FE3}	103	4 V	$\overline{V_{S2}}$														"	18 to 14	E8	V	$h_{FE3} = \frac{E8}{1000}$	See figures 8 and 9 for test circuit and waveforms	10	-	-
	h_{FE3}	104	4 V	"														3	"	E9	"	$h_{FE3} = \frac{E9}{1000}$	"	10	-	-
	h_{FE4}	105	6 V	"														None	"	E10	"	$h_{FE4} = \frac{E10}{3000}$	"	15	-	-
	h_{FE4}	106	6 V	"														3	"	E11	"	$h_{FE4} = \frac{E11}{3000}$	"	15	-	-
	V_{BE1}	107	4 V	"	$\overline{V_{S3}}$													1,2	12 to 16	E12	"	V_{BE1} - E12	"	-	1.2	V
	V_{BE1}	108	4 V	"	"													1,2,3	11 to 10	E13	"	V_{BE1} - E13	"	-	1.2	"
	V_{BE2}	109	6 V	"	"													1,2	12 to 16	E14	"	V_{BE2} - E14	"	-	1.4	"
	V_{BE2}	110	6 V	"	"													1,2,3	11 to 10	E15	"	V_{BE2} - E15	"	-	1.4	"
	V_{CESAT1}	111	4 V	"	"													1,2	5 to 6	E16	"	V_{CESAT1} - E16	"	-	0.5	"
	V_{CESAT1}	112	4 V	"	"													1,2,3	10 to 9	E17	"	V_{CESAT1} - E17	"	-	0.5	"
	V_{CESAT2}	113	6 V	"	"													1,2	5 to 6	E18	"	V_{CESAT2} - E18	"	-	0.8	"
	V_{CESAT2}	114	6 V	"	"													1,2,3	10 to 9	E19	"	V_{CESAT2} - E19	"	-	0.8	"
	V_{CBO1}	115				GND	100 µA											3	15 to 12	E20	"	V_{CBO1} - E20	See figure 8 for test circuit	35	-	"
	V_{CBO1}	116							GND		100 µA							4	5 to 3	E21	"	V_{CBO1} - E21	"	35	-	"
	V_{CER1}	117					100 µA	GND										4	15 to 16	E22	"	V_{CER1} - E22	"	35	-	"
	V_{CER1}	118								GND	100 µA							3,4	5 to 3	E23	"	V_{CER1} - E23	"	35	-	"
	V_{EBO}	119				GND		100 µA										3,4	16 to 12	E24	"	V_{EBO} - E24	"	5	-	"
	V_{EBO}	120							GND	100 µA								4	3 to 4	E25	"	V_{EBO} - E25	"	5	-	"

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroups	Symbol	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Reserved terminal	Note	Test limits		Unit
			1A	1B	1C	1D	1E	GND	1F	1G	1K	1L	2A	2B	2C	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH1}	121	3 V														5	See figure 13 for test circuit and waveforms		15	ns
		122											A				10				
		123													B		6				
		124											A				10			20	
		125													B		5				
		126											A				10			20	
		127													B		6			15	
		128											A				10			15	
		129												OUT		4.5 V	2 to 3	See figure 11 for test circuit and waveforms		30	
		130												OUT			1 to 3				
		131	3 V	IN													13 to 12				
		132												OUT			1 to 12				
		133	3 V	IN													2 to 3			15	
		134												OUT			1 to 3				
		135	3 V	IN													13 to 12				
		136												OUT			1 to 12				
		137	3 V	IN													2 to 5	See figure 12 for test circuit and waveforms		30	
		138													OUT		1 to 5				
		139	3 V	IN													13 to 10				
		140													OUT		1 to 10				
		141	3 V	IN													2 to 5				
		142													OUT		1 to 5				
		143	3 V	IN													13 to 10				
		144													OUT		1 to 10				
		145	3 V	IN													6			15	
		146	3 V	IN													10				
		147	3 V	IN													6				
		148	3 V	IN													10				

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroups	Symbol	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Note	Measured terminal	Test limits		Unit
																			Min	Max	
10 T_A = 125°C	t_{PLH1}	149			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	5	-	22.5	ns
	t_{PHL1}	150			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	10	-	22.5	ns
	t_{PLH1}	151			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	5	-	30	ns
	t_{PHL1}	152			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	10	-	30	ns
	t_{PLH1}	153			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	5	-	22.5	ns
	t_{PHL1}	154			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	10	-	22.5	ns
	t_{PLH1}	155			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	5	-	22.5	ns
	t_{PHL1}	156			A	B		GND		GND	CKO	B	A				See figure 13 for test circuit and waveforms	10	-	22.5	ns
	t_{PLH1}	157	3 V	IN			OUT									4.5 V	See figure 11 for test circuit and waveforms	2 to 3	-	45	ns
		158	3 V	IN			OUT											1 to 3	-		ns
		159	3 V	IN										OUT	IN	3 V		13 to 12	-	22.5	ns
		160	3 V	IN										OUT	IN	3 V		1 to 12	-		ns
	t_{PHL1}	161	3 V	IN			OUT											2 to 3	-		ns
		162	3 V	IN			OUT											1 to 3	-		ns
		163	3 V	IN										OUT	IN	3 V		13 to 12	-		ns
		164	3 V	IN										OUT	IN	3 V		1 to 12	-		ns
	t_{PLH2}	165	3 V	IN			Connect to 18					OUT					See figure 12 for test circuit and waveforms	2 to 5	-	45	ns
		166	3 V	IN			Connect to 18					OUT						1 to 5	-		ns
		167	3 V	IN								OUT		Connect to 20	IN	3 V		13 to 10	-		ns
		168	3 V	IN								OUT		Connect to 20	IN	3 V		1 to 10	-		ns
	t_{PHL2}	169	3 V	IN			Connect to 18					OUT						2 to 5	-		ns
		170	3 V	IN			Connect to 18					OUT						1 to 5	-		ns
		171	3 V	IN								OUT		Connect to 20	IN	3 V		13 to 10	-		ns
		172	3 V	IN								OUT		Connect to 20	IN	3 V		1 to 10	-		ns
	t_{TLH1}	173	3 V	IN			Connect to 18					OUT						5	-	22.5	ns
	t_{THL1}	174	3 V	IN								OUT		Connect to 20	IN			10	-		ns
	t_{TLH1}	175	3 V	IN			Connect to 18					OUT						5	-		ns
	t_{THL1}	176	3 V	IN								OUT		Connect to 20	IN			10	-		ns

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Note	Measured terminal	Test limits Min	Test limits Max	Unit
			1A	1B	1C	1D	1E	GND	2A	2B	2C	2D	2E	2Y	2Z	VCC					
11 T_A = -55°C	tPLH1	177					GND						A				See figure 13 for test circuit and waveforms	5	-	22.5	ns
	tPHL1	178					GND						A					10	-	-	
	tPLH2	179					GND						A					5	-	-	
	tPHL2	180					GND						A					10	-	-	
	tTLH1	181					GND						A					5	-	30	
	tTHL1	182					GND						A					10	-	30	
	tTLH2	183					GND						A					5	-	22.5	
	tTHL2	184					GND						A					10	-	22.5	
	tPLH1	185		OUT												4.5 V	See figure 11 for test circuit and waveforms	2 to 3	-	45	
		186		OUT														1 to 3	-	-	
	tPHL1	187		OUT														13 to 12	-	22.5	
		188		OUT														1 to 12	-	-	
	tPLH2	189		OUT														2 to 3	-	45	
		190		OUT														1 to 3	-	-	
	tPHL2	191		OUT														13 to 12	-	-	
		192		OUT														1 to 12	-	-	
	tTLH1	193		Connect to 1B							OUT						See figure 12 for test circuit and waveforms	2 to 5	-	45	
	tTHL1	194		Connect to 1B							OUT							1 to 5	-	-	
	tTLH2	195		Connect to 1B														13 to 10	-	-	
		196		Connect to 1B														1 to 10	-	-	
	tTHL2	197		Connect to 1B							OUT							2 to 5	-	-	
		198		Connect to 1B							OUT							1 to 5	-	-	
	tTLH1	199		Connect to 1B														13 to 10	-	-	
		200		Connect to 1B														1 to 10	-	-	
	tTHL1	201		Connect to 1B							OUT							5	-	22.5	
	tTLH2	202		Connect to 1B														10	-	-	
	tTHL2	203		Connect to 1B							OUT							5	-	-	
		204		Connect to 1B														10	-	-	

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TABLE III. Group A inspection for device type 02.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"	"	2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	mA
	"	14	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	16				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	17	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL2}	18	GND	GND		"		GND	GND	"	"	8	-	65	"
	V_{OL1}	19	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL1}	20				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL5}	21	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL5}	22				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH1}	23	2 V	2 V	30 V	"				"	"	3	-	300	uA
	I_{OH1}	24				"	30 V	2 V	2 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
$T_A = 125°C$ [2]	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	µA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	38	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	40				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	41	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL2}	42	GND	GND		"		GND	GND	"	"	8	-	65	"
	V_{OL1}	43	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL1}	44				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL5}	45	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL5}	46				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH1}	47	2 V	2 V	30 V	"				"	"	3	-	300	µA
	I_{OH1}	48				"	30 V	2 V	2 V	"	"	5	-	300	µA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
$T_A = -55°C$ [3]	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"	"	2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	62	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	64				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	65	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL2}	66	GND	GND		"		GND	GND	"	"	8	-	65	"
	V_{OL1}	67	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL1}	68				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL5}	69	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL5}	70				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH1}	71	2 V	2 V	30 V	"				"	"	3	-	300	uA
	I_{OH1}	72				"	30 V	2 V	2 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	3 V	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	30	ns
	"	74	3 V	IN	OUT	"				"		2 to 3	-	"	"
	"	75				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	76				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	78	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	80				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	14	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	14	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	20	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	20	"
10 T_A = 125°C	t_{PLH}	85	IN	3 V	OUT	"				"	"	1 to 3	-	45	"
	"	86	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	88				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	90	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	92				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	3 V	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	45	ns
	"	98	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	99				"	OUT	IN	3 V	"	"	5 to 5	-	"	"
	"	100				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	102	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	104				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
T_A = 25°C [1]	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		-				-	-	2 to 4	-	-	-
	"	3				-		-12 mA	4.5 V	-	-	6 to 4	-	-	-
	"	4				-		4.5 V	-12 mA	-	-	7 to 4	-	-	-
	I_{IH1}	5	2.4 V	GND		-				5.5 V	-	1 to 4	-	40	uA
	"	6	GND	2.4 V		-				-	-	2 to 4	-	-	-
	"	7				-		2.4 V	GND	-	-	6 to 4	-	-	-
	"	8				-		GND	2.4 V	-	-	7 to 4	-	-	-
	I_{IH2}	9	5.5 V	GND		-				-	-	1 to 4	-	1	mA
	"	10	GND	5.5 V		-				-	-	2 to 4	-	-	-
	"	11				-		5.5 V	GND	-	-	6 to 4	-	-	-
	"	12				-		GND	5.5 V	-	-	7 to 4	-	-	-
	I_{IL1}	13	0.4 V	5.5 V		-				-	-	1 to 4	-1.6	-	-
	"	14	5.5 V	0.4 V		-				-	-	2 to 4	-	-	-
	"	15				-		0.4 V	5.5 V	-	-	6 to 4	-	-	-
	"	16				-		5.5 V	0.4 V	-	-	7 to 4	-	-	-
	I_{CCH}	17	GND	GND		-		GND	GND	-	-	8	-	14	-
	I_{CCL}	18	5.5 V	5.5 V		-		5.5 V	5.5 V	-	-	8	-	71	-
	V_{OL2}	19	2 V	2 V	100 mA	-	100 mA	2 V	2 V	4.5 V	-	3 to 4	-	0.5	V
	V_{OL2}	20				-				-	-	5 to 4	-	0.5	-
	V_{OL6}	21	2 V	2 V	300 mA	-	300 mA	2 V	2 V	-	-	3 to 4	-	0.8	-
	V_{OL6}	22				-				-	-	5 to 4	-	0.8	-
	I_{OH2}	23	0.8 V	4.5 V	30 V	-	30 V	0.8 V	4.5 V	-	-	3	-	300	uA
	I_{OH2}	24				-				-	-	5	-	300	uA

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			MIN	MAX	
$T_A = 125°C$ [2]	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"	circuit	2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	38	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	40				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH4}	41	GND	GND		"		GND	GND	"	"	8	-	14	"
	I_{CCL4}	42	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	71	"
	V_{OL2}	43	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL2}	44				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL6}	45	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL6}	46				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH2}	47	0.8 V	4.5 V	30 V	"				"	"	3	-	300	uA
	I_{OH2}	48				"	30 V	0.8 V	4.5 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
[3] T_A = -55°C	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	µA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	62	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	64				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH4}	65	GND	GND		"		GND	GND	"	"	8	-	14	"
	I_{CCL4}	66	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	71	"
	V_{OL2}	67	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL2}	68				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL6}	69	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL6}	70				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH2}	71	0.8 V	4.5 V	30 V	"				"	"	3	-	300	µA
	I_{OH2}	72				"	30 V	0.8 V	4.5 V	"	"	5	-	300	µA

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	3 V	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	35	ns
	"	74	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	75				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	76				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	78	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	80				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	14	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	14	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	20	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	20	"
10 T_A = 125°C	t_{PLH}	85	IN	3 V	OUT	"				"	"	1 to 3	-	55	"
	"	86	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	88				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	90	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	92				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	1 1A	2 1B	3 1Y	4 GND	5 2Y	6 2A	7 2B	8 V_{CC}	Note	Measured terminal	Test limits Min	Test limits Max	Unit
11 T_A = -55°C	t_{PLH}	97	IN	3 V	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	55	ns
	"	98	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	99				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	100				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	102	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	104				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	18.5	"
	"	106				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	"	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 04.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	14	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	16				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	17	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL3}	18	GND	GND		"		GND	GND	"	"	8	-	68	"
	V_{OL3}	19	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL3}	20				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL7}	21	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL7}	22				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH3}	23	2 V	GND	30 V	"				"	"	3	-	300	uA
	I_{OH3}	24				"	30 V	2 V	GND	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
$T_A = 125°C$ [2]	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
"	"	26	4.5 V	-12 mA		"				"	"	2 to 4	-	"	"
"	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
"	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	µA
"	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
"	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
"	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
"	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
"	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
"	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
"	"	38	GND	0.4 V		"				"	"	2 to 4	-	"	"
"	"	39				"		0.4 V	GND	"	"	6 to 4	-	"	"
"	"	40				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	41	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL2}	42	GND	GND		"		GND	GND	"	"	8	-	65	"
	V_{OL3}	43	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL3}	44				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL7}	45	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL7}	46				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH3}	47	2 V	GND	30 V	"				"	"	3	-	300	µA
	I_{OH3}	48				"	30 V	2 V	GND	"	"	5	-	300	µA

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
$T_A = -55°C$ [3]	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.8 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	62	GND	0.8 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	64				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	65	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL2}	66	GND	GND		"		GND	GND	"	"	8	-	65	"
	V_{OL3}	67	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL3}	68				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL7}	69	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL7}	70				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH3}	71	2 V	GND	30 V	"				"	"	3	-	300	uA
	I_{OH3}	72				"	30 V	2 V	GND	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	GND	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	30	ns
	"	74	GND	IN	OUT	"				"		2 to 3	-	"	"
	"	75				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	76				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	GND	OUT	"				"	"	1 to 3	-	"	"
	"	78	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	80				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	14	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	14	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	20	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	20	"
10 T_A = 25°C	t_{PLH}	85	IN	GND	OUT	"				"	"	1 to 3	-	45	"
	"	86	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	88				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	GND	OUT	"				"	"	1 to 3	-	"	"
	"	90	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	GND	"	"	6 to 9	-	"	"
	"	92				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	GND	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	45	ns
	"	98	GND	IN	OUT	"				"		2 to 3	-	"	"
	"	99				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	100				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	GND	OUT	"				"	"	1 to 3	-	"	"
	"	102	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	104				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 05.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	14	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	16				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH3}	17	GND	GND		"		GND	GND	"	"	8	-	17	"
	I_{CCL7}	18	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	79	"
	V_{OL4}	19	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL4}	20				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	21	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	22				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH4}	23	0.8 V	0.8 V	30 V	"				"	"	3	-	300	uA
	I_{OH4}	24				"	30 V	0.8 V	0.8 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
2 T_A = 125°C	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	38	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	40				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH3}	41	GND	GND		"		GND	GND	"	"	8	-	17	"
	I_{CCL7}	42	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	79	"
	V_{OL4}	43	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL4}	44				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	45	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	46				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH4}	47	0.8 V	0.8 V	30 V	"				"	"	3	-	300	uA
	I_{OH4}	48				"	30 V	0.8 V	0.8 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
3 T_A = -55°C	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	62	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	64				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH3}	65	GND	GND		"		GND	GND	"	"	8	-	17	"
	I_{CCL7}	66	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	79	"
	V_{OL4}	67	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL4}	68				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	69	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	70				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH4}	71	0.8 V	0.8 V	30 V	"				"	"	3	-	300	uA
	I_{OH4}	72				"	30 V	0.8 V	0.8 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	GND	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	35	ns
	"	74	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	75				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	76				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	GND	OUT	"				"	"	1 to 3	-	"	"
	"	78	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	80				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	14	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	14	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	20	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	20	"
10 T_A = 125°C	t_{PLH}	85	IN	GND	OUT	"				"	"	1 to 3	-	55	"
	"	86	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	88				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	GND	OUT	"				"	"	1 to 3	-	"	"
	"	90	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	92				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	GND	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	55	ns
	"	98	GND	IN	OUT	"				"		2 to 3	-	"	"
	"	99				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	100				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	GND	OUT	"				"	"	1 to 3	-	"	"
	"	102	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	104				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	18.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	18.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 06.

Subgroup	Symbol	Test no.	1	2	17	12	18	16	4	3	5	8	9	10	11	7	6	Relays energized	Measured terminal			Equations	Note	Test limits		Unit
			V_{S1}	V_{S2}	V_{S3}	1B	1C	1E	2B	2E	2C	V_{CC}	G	1A	1Y	2A	2Y		No.	Value	UNIT			Min	Max	
T_A = 25°C	V_{IC}	1										4.5 V	-12 mA	4.5 V		4.5 V		None	9	E1	V	V_{IC} = E1	See figure 8 for test circuit	-	-1.5	V
	V_{IC}	2										4.5 V	4.5 V	-12 mA		4.5 V		"	10	E2	V	V_{IC} = E2	"	-	-1.5	V
	V_{IC}	3										4.5 V	4.5 V	4.5 V		-12 mA		"	7	E3	V	V_{IC} = E3	"	-	-1.5	V
	I_{IH1}	4										5.5 V	GND	2.4 V		GND		"	10	I1	A	I_{IH1} = I1	"	-	40	µA
	I_{IH1}	5										"	GND	GND		2.4 V		"	7	I2	"	I_{IH1} = I2	"	-	40	µA
	I_{IH2}	6										"	GND	5.5 V		GND		"	10	I3	"	I_{IH2} = I3	"	-	1	mA
	I_{IH2}	7										"	GND	GND		5.5 V		"	7	I4	"	I_{IH2} = I4	"	-	1	mA
	I_{IH3}	8										"	2.4 V	GND		GND		"	9	I5	"	I_{IH3} = I5	"	-	80	µA
	I_{IH4}	9										"	5.5 V	GND		GND		"	9	I6	"	I_{IH4} = I6	"	-	2	mA
	I_{IL1}	10										"	5.5 V	0.4 V		5.5 V		"	10	I7	"	I_{IL1} = I7	"	-	-1.6	"
	I_{IL1}	11										"	5.5 V	5.5 V		0.4 V		"	7	I8	"	I_{IL1} = I8	"	-	-1.6	"
	I_{IL2}	12										"	0.4 V	5.5 V		5.5 V		"	9	I9	"	I_{IL2} = I9	"	-	-3.2	"
	I_{CCH1}	13										"	GND	GND		GND		"	8	I10	"	I_{CCH1} = I10	"	-	4	"
	I_{CCL1}	14										"	5.5 V	5.5 V		5.5 V		"	8	I11	"	I_{CCL1} = I11	"	-	11	"
	V_{OL9}	15										4.5 V	2 V	2 V	I_{OL} = 16 mA			"	11	E4	V	V_{OL9} = E4	"	-	0.5	V
	V_{OL9}	16										"	2 V			2 V	I_{OL} = -16 mA	"	6	E5	"	V_{OL9} = E5	"	-	0.5	"
	V_{OH}	17										"	4.5 V	0.8 V	I_{OH} = -400 µA			"	11	E6	"	V_{OH} = E6	"	2.4	-	"
	V_{OH}	18										"	4.5 V			0.8 V	I_{OH} = -400 µA	"	6	E7	"	V_{OH} = E7	"	2.4	-	"
	I_{OS1}	19										5.5 V	GND	GND	GND			"	11	I12	A	I_{OS1} = I12	"	-	-55	mA
	I_{OS1}	20										5.5 V	"			GND	GND	"	6	I13	"	I_{OS1} = I13	"	-	-55	"
	I_{OS2}	21										4.5 V	"	GND	GND			"	11	I14	"	I_{OS2} = I14	"	-18	-	"
	I_{OS2}	22										4.5 V	"			GND	GND	"	6	I15	"	I_{OS2} = I15	"	-18	-	"
	h_{FE1}	23	4 V	$\overline{V_{S2}}$														"	10 to 14	E8	V	$h_{FE1} = \frac{E8}{1000}$	See figures 8 and 9 for test circuit and waveforms	25	-	-
	h_{FE1}	24	4 V	"														3	"	E9	"	$h_{FE1} = \frac{E9}{1000}$		25	-	-
	h_{FE2}	25	6 V	"														None	"	E10	"	$h_{FE2} = \frac{E10}{3000}$	"	30	-	-
	h_{FE2}	26	6 V	"														3	"	E11	"	$h_{FE2} = \frac{E11}{3000}$	"	30	-	-
	V_{BE1}	27	4 V	"	$\overline{V_{S3}}$													1,2	12 to 16	E12	"	V_{BE1} = E12	"	-	1.7	V
	V_{BE1}	28	4 V	"	$\overline{V_{S3}}$													1,2,3	14 to 3	E13	"	V_{BE1} = E13	"	-	1.7	V

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	13	16	4	3	5	8	9	11	14	7	6	Relays energized	Measured terminal No.	Measured terminal Value	Measured terminal Unit	Equations	Note	Test limits Min	Test limits Max	Unit
			V_{S1}	V_{S2}	V_{S3}	1B	1C	1E	2B	2E	2C	V_{CC}	G	1K	1Y	2A	2Y									
T_A = 25°C [1]	V_{BE2}	29	6 V	$\overline{V_{S2}}$	$\overline{V_{S3}}$													1,2	12 to 16	E14	V	V_{BE2} = E14	See figures 8 and 9 for test circuit and waveforms	-	1.4	V
	V_{BE2}	30	6 V															1,2,3	4 to 3	E15	"	V_{BE2} = E15		-	1.4	"
	V_{CESAT1}	31	4 V															1,2	15 to 16	E16	"	V_{CESAT1} = E16	"	-	0.5	"
	V_{CESAT1}	32	4 V															1,2,3	5 to 3	E17	"	V_{CESAT1} = E17	"	-	0.5	"
	V_{CESAT2}	33	6 V															1,2	15 to 16	E18	"	V_{CESAT2} = E18	"	-	0.8	"
	V_{CESAT2}	34	6 V															1,2,3	5 to 3	E19	"	V_{CESAT2} = E19	"	-	0.8	"
	V_{CBO2}	35				GND	100 µA											3	15 to 12	E20	"	V_{CBO2} = E20	"	40	-	"
	V_{CBO2}	36							GND		100 µA							4	5 to 4	E21	"	V_{CBO2} = E21	"	40	-	"
	V_{CER2}	37					100 µA	GND										4	15 to 16	E22	"	V_{CER1} = E22	"	40	-	"
	V_{CER2}	38								GND	100 µA							3,4	5 to 3	E23	"	V_{CER1} = E23	"	40	-	"
	V_{CEO1}	39					10 mA	GND										3	15 to 16	E24	"	V_{CEO1} = E24	"	25	-	"
	V_{CEO1}	40								GND	10 mA							3,4	5 to 3	E25	"	V_{CEO1} = E25	"	25	-	"
	V_{EBO}	41				GND		100 µA										3,4	16 to 12	E26	"	V_{EBO} = E26	"	5	-	"
	V_{EBO}	42							GND		100 µA							4	3 to 4	E27	"	V_{EBO} = E27	"	5	-	"
T_A = 125°C [2]	V_{IC}	43										4.5 V	-12 mA	4.5 V		4.5 V		None	9	E1	"	V_{IC} = E1	See figure 8 for test circuit	-	-1.5	"
	V_{IC}	44										4.5 V	4.5 V	-12 mA		4.5 V		"	10	E2	"	V_{IC} = E2	"	-	-1.5	"
	V_{IC}	45										4.5 V	4.5 V	4.5 V		-12 mA		"	7	E3	"	V_{IC} = E3	"	-	-1.5	"
	I_{IH1}	46										5.5 V	GND	2.4 V		GND		"	10	I1	A	I_{IH1} = I1	"	-	40	µA
	I_{IH1}	47										"	GND	GND		2.4 V		"	7	I2	"	I_{IH1} = I2	"	-	40	µA
	I_{IH2}	48										"	GND	5.5 V		GND		"	10	I3	"	I_{IH2} = I3	"	-	1	mA
	I_{IH2}	49										"	GND	GND		5.5 V		"	7	I4	"	I_{IH2} = I4	"	-	1	mA
	I_{IH3}	50										"	2.4 V	GND		GND		"	9	I5	"	I_{IH3} = I5	"	-	80	µA
	I_{IH4}	51										"	5.5 V	GND		GND		"	9	I6	"	I_{IH4} = I6	"	-	2	mA
	I_{IL1}	52										"	4.5 V	0.4 V		4.5 V		"	10	I7	"	I_{IL1} = I7	"	-	-1.6	"
	I_{IL1}	53										"	4.5 V	4.5 V		0.4 V		"	7	I8	"	I_{IL1} = I8	"	-	-1.6	"
	I_{IL2}	54										"	0.4 V	4.5 V		4.5 V		"	9	I9	"	I_{IL2} = I9	"	-	-3.2	"
	I_{CCH1}	55										"	GND	GND		GND		"	8	I10	"	I_{CCH1} = I10	"	-	4	"
	I_{CCL1}	56										"	5.5 V	5.5 V		5.5 V		"	8	I11	"	I_{CCL1} = I11	"	-	11	"
	V_{OL9}	57										4.5 V	2 V	2 V	I_{OL} = 16 mA			"	11	E4	V	V_{OL9} = E4	"	-	0.5	V
	V_{OL9}	58										"	2 V			2 V	I_{OL} = 16 mA	"	6	E5	"	V_{OL9} = E5	"	-	0.5	"
	V_{OH}	59										"	4.5 V	0.8 V	I_{OH} = -400 µA			"	11	E6	"	V_{OH} = E6	"	2.4	-	"
	V_{OH}	60										"	4.5 V			0.8 V	I_{OH} = -400 µA	"	6	E7	"	V_{OH} = E7	"	2.4	-	"

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	15	16	4	3	5	8	9	10	11	7	6	Relays ener-gized	Measured terminal No.	Measured terminal VBYBO	Measured terminal Unit	Equations	Note	Test limits Min	Test limits Max	Unit
			V_{S1}	V_{S2}	V_{S3}	2B	2C	2E	2B	2C	2C	V_{CC}	G	1A	1Y	2A	2Y									
T_A = 125°C 2	I_{OS1}	61										5.5 V	GND	GND	GND			None	11	I12	A	I_{OS1} = I12	See figure 8 for test	-	-55	mA
	I_{OS1}	62										5.5 V	"			GND	GND	"	6	I13	"	I_{OS1} = I13	circuit	-	-55	"
	I_{OS2}	63										4.5 V	"	GND	GND			"	11	I14	"	I_{OS2} = I14	"	-18	-	"
	I_{OS2}	64										4.5 V	"			GND	GND	"	6	I15	"	I_{OS2} = I15	"	-18	-	"
	h_{FE1}	65	4 V	$\overline{V_{S2}}$														"	18 to 14	E8	V	$h_{FE1} = \frac{E8}{1000}$	See figures 8 and 9 for test circuit and waveforms	25	-	-
	h_{FE1}	66	4 V	"														3	"	E9	"	$h_{FE1} = \frac{E9}{1000}$		25	-	-
	h_{FE2}	67	6 V	"														None	"	E10	"	$h_{FE2} = \frac{E10}{3000}$	"	30	-	-
	h_{FE2}	68	6 V	"														3	"	E11	"	$h_{FE3} = \frac{E11}{3000}$	"	30	-	-
	V_{BE1}	69	4 V	"	$\overline{V_{S3}}$													1,2	12 to 16	E12	"	V_{BE1} = E12	"	-	1.2	V
	V_{BE1}	70	4 V	"	"													1,2,3	4 to 3	E13	"	V_{BE1} = E13	"	-	1.2	"
	V_{BE2}	71	6 V	"	"													1,2	12 to 16	E14	"	V_{BE2} = E14	"	-	1.4	"
	V_{BE2}	72	6 V	"	"													1,2,3	4 to 3	E15	"	V_{BE2} = E15	"	-	1.4	"
	V_{CESAT1}	73	4 V	"	"													1,2	15 to 16	E16	"	V_{CESAT1} = E16	"	-	0.5	"
	V_{CESAT1}	74	4 V	"	"													1,2,3	5 to 3	E17	"	V_{CESAT1} = E17	"	-	0.5	"
	V_{CESAT2}	75	6 V	"	"													1,2	15 to 16	E18	"	V_{CESAT2} = E18	"	-	0.8	"
	V_{CESAT2}	76	6 V	"	"													1,2,3	5 to 3	E19	"	V_{CESAT2} = E19	"	-	0.8	"
	V_{CBO2}	77				GND	100 µA											3	15 to 12	E20	"	V_{CBO2} = E20	"	40	-	"
	V_{CBO2}	78							GND		100 µA							4	5 to 4	E21	"	V_{CBO2} = E21	"	40	-	"
	V_{CER2}	79					100 µA	GND										4	15 to 16	E22	"	V_{CER1} = E22	"	40	-	"
	V_{CER2}	80								GND	100 µA							3,4	5 to 3	E23	"	V_{CER1} = E23	"	40	-	"
	V_{CEO1}	81					10 mA	GND										3	15 to 16	E24	"	V_{CEO1} = E24	"	25	-	"
	V_{CEO1}	82								GND	10 mA							3,4	5 to 3	E25	"	V_{CEO1} = E25	"	25	-	"
	V_{EBO}	83				GND		100 µA										3,4	16 to 12	E26	"	V_{EBO} = E26	"	5	-	"
	V_{EBO}	84							GND		100 µA							4	3 to 4	E27	"	V_{EBO} = E27	"	5	-	"
T_A = -55°C 3	V_{IC}	85										4.5 V	-12 mA	4.5 V		4.5 V		None	9	E1	"	V_{IC} = E1	See figure 8 for test	-	-1.5	"
	V_{IC}	86										4.5 V	4.5 V	-12 mA		4.5 V		"	10	E2	"	V_{IC} = E2	circuit	-	-1.5	"
	V_{IC}	87										4.5 V	4.5 V	4.5 V		-12 mA		"	7	E3	"	V_{IC} = E3	"	-	-1.5	"
	I_{IH1}	88										5.5 V	GND	2.4 V		GND		"	10	I1	A	I_{IH1} = I1	"	-	40	µA
	I_{IH1}	89											GND	GND		2.4 V		"	7	I2	"	I_{IH1} = I2	"	-	40	µA
	I_{IH2}	90											GND	5.5 V		GND		"	10	I3	"	I_{IH2} = I3	"	-	1	mA
	I_{IH2}	91											GND	GND		5.5 V		"	7	I4	"	I_{IH2} = I4	"	-	1	mA
	I_{IH3}	92											2.4 V	GND		GND		"	9	I5	"	I_{IH3} = I5	"	-	80	µA
	I_{IH4}	93											5.5 V	GND		GND		"	9	I6	"	I_{IH4} = I6	"	-	2	mA

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	13	14	4	3	5	8	9	10	11	7	6	Relays energized	Measured terminal			Equations	Note	Test limits		Unit
																			No.	Value	Unit			Min	Max	
$T_A = -55°C$ [3]	I_{IL1}	94										5.5 V	4.5 V	0.4 V		4.5 V		None	10	I7	A	I_{IL1} - I7	See figure 8 for test circuit	-	-1.6	mA
	I_{IL1}	95										"	4.5 V	4.5 V		0.4 V		"	7	I8	"	I_{IL1} - I8	"	-	-1.6	"
	I_{IL2}	96										"	0.4 V	4.5 V		4.5 V		"	9	I9	"	I_{IL2} - I9	"	-	-3.2	"
	I_{CCH1}	97										"	GND	GND		GND		"	8	I10	"	I_{CCH1} - I10	"	-	4	"
	I_{CCL1}	98										"	5.5 V	5.5 V		5.5 V		"	8	I11	"	I_{CCL1} - I11	"	-	11	"
	V_{OL9}	99										4.5 V	2 V	2 V	I_{OL} = 16 mA			"	11	E4	V	V_{OL9} - E4	"	-	0.5	V
	V_{OL9}	100										"	2 V			2 V	I_{OL} = 16 mA	"	6	E5	"	V_{OL9} - E5	"	-	0.5	"
	V_{OH}	101										"	4.5 V	0.8 V	I_{OH} = -400 µA			"	11	E6	"	V_{OH} - E6	"	2.4	-	"
	V_{OH}	102										"	4.5 V			0.8 V	I_{OH} = -400 µA	"	6	E7		V_{OH} - E7	"	2.4	-	"
	I_{OS1}	103										5.5 V	GND	GND	GND			"	11	I12	A	I_{OS1} - I12	"	-	-55	mA
	I_{OS1}	104										5.5 V	"			GND	GND	"	6	I13	"	I_{OS1} - I13	"	-	-55	"
	I_{OS2}	105										4.5 V	"	GND	GND			"	11	I14	"	I_{OS2} - I14	"	-18	-	"
	I_{OS2}	106										4.5 V	"			GND	GND	"	6	I15	"	I_{OS2} - I15	"	-18	-	"
	h_{FE3}	107	6 V	$\overline{S2}$														"	18 to 14	E8	V	$h_{FE3} = \frac{E8}{1000}$	See figures 8 and 9 for test circuit and waveforms	10	-	-
	h_{FE3}	108	4 V	"														3	"	E9	"	$h_{FE3} = \frac{E9}{1000}$		10	-	-
	h_{FE4}	109	6 V	"														None	"	E10	"	$h_{FE3} = \frac{E10}{3000}$	"	15	-	-
	h_{FE4}	110	6 V	"														3	"	E11	"	$h_{FE4} = \frac{E11}{3000}$	"	15	-	-
	V_{BE1}	111	4 V	"	$\overline{S3}$													1,2	12 to 16	E12	"	V_{BE1} - E12	"	-	1.2	V
	V_{BE1}	112	4 V	"	"													1,2,3	4 to 3	E13	"	V_{BE1} - E13	"	-	1.2	"
	V_{BE2}	113	6 V	"	"													1,2	12 to 16	E14	"	V_{BE2} - E14	"	-	1.4	"
	V_{BE2}	114	6 V	"	"													1,2,3	4 to 3	E15	"	V_{BE2} - E15	"	-	1.4	"
	V_{CESAT1}	115	4 V	"	"													1,2	15 to 16	E16	"	V_{CESAT1} - E16	"	-	0.5	"
	V_{CESAT1}	116	4 V	"	"													1,2,3	5 to 3	E17	"	V_{CESAT1} - E17	"	-	0.5	"
	V_{CESAT2}	117	6 V	"	"													1,2	15 to 16	E18	"	V_{CESAT2} - E19	"	-	0.8	"
	V_{CESAT2}	118	6 V	"	"													1,2,3	5 to 3	E19	"	V_{CESAT2} - E19	"	-	0.8	"
	V_{CBO2}	119		"	"	GND	100 µA											3	15 to 12	E20	"	V_{CBO2} - E20	"	40	-	"
	V_{CBO2}	120		"	"				GND		100 µA							4	5 to 4	E21	"	V_{CBO2} - E21	"	40	-	"
	V_{CER2}	121		"	"		100 µA	GND										4	15 to 16	E22	"	V_{CER1} - E22	"	40	-	"
	V_{CER2}	122		"	"					GND	100 µA							3,4	5 to 3	E23	"	V_{CER1} - E23	"	40	-	"

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	1	2	17	12	15	18	4	3	5	8	9	10	11	7	6	Measured terminal			Equations	Note	Test limits		Unit
																		Logic level	Meas.	Voltage			Min	Max	
3 T_A = -55°C	V_{CEO1}	123	V_{S1}	V_{S2}	V_{S3}		10 mA	GND		GND	10 mA							3	15 to 16	E24	V	V_{CEO1} - E24	See figures 8 (and 9 for test circuit and waveforms	25		V
	V_{CEO1}	124																3,4	15 to 3	E25	-	V_{CEO1} - E25		25		-
	V_{CEO}	125				GND			100 µA									3,4	16 to 12	E26	-	V_{CEO} - E26	-	5		-
	V_{CEO}	126								GND	100 µA							4	13 to 4	E27	-	V_{CEO} - E27		5		-

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroups	Symbol	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Note	Measured terminal	Test limits Min	Test limits Max	Unit
			B	IIL	IIH	1Y	1C	1E	GND	Sub	2K	2C	2A	2Y	3A	Vcc					
T_A = 25°C	t_{PD}	127					B	GND	GND	GND	GND	B	A				See figure 13 for test circuit and waveforms	5		15	ns
	t_{PD}	128					B	GND	GND	GND	GND	B	A					10		15	
	t_{CS}	129					B	GND	GND	GND	GND	B	A					5		23	
	t_{CS}	130					B	GND	GND	GND	GND	B	A					10		23	
	t_{CA}	131					B	GND	GND	GND	GND	B	A					5		20	
	t_{CA}	132					B	GND	GND	GND	GND	B	A					10		20	
	t_{CF}	133					B	GND	GND	GND	GND	B	A					5		15	
	t_{CF}	134					B	GND	GND	GND	GND	B	A					10		15	
	t_{PLH1}	135	3 V	IIH		OUT										0.5 V	See figure 11 for test circuit and waveforms	2 to 3		90	
		136	IIH	3 V		OUT												11 to 3			
		137	3 V	IIH										OUT	IIH			13 to 12			
		138	IIH	3 V										OUT	3 V			1 to 12			
	t_{PHL1}	139	3 V	IIH		OUT												2 to 3		15	
		140	IIH	3 V		OUT												11 to 3			
		141	3 V	IIH										OUT	IIH			13 to 12			
		142	IIH	3 V										OUT	3 V			11 to 12			
	t_{PLH3}	143	3 V	IIH		Connect to 1B	OUT										See figure 12 for test circuit and waveforms	2 to 5		65	
		144	IIH	3 V		Connect to 1B	OUT											11 to 5			
		145	3 V	IIH								OUT	IIH	[Connect to 2B				13 to 10			
		146	IIH	3 V								OUT	3 V	[Connect to 2B				11 to 10			
	t_{PHL3}	147	3 V	IIH		Connect to 1B	OUT											2 to 5		50	
		148	IIH	3 V		Connect to 1B	OUT											11 to 5			
		149	3 V	IIH								OUT	IIH	[Connect to 2B				13 to 10			
		150	IIH	3 V								OUT	3 V	[Connect to 2B				11 to 10			
	t_{TLH2}	151	3 V	IIH		Connect to 1B	OUT											5		20	
	t_{TLH2}	152	3 V	IIH								OUT	IIH	[Connect to 2B				10			
	t_{THL2}	153	3 V	IIH		Connect to 1B	OUT											5			
	t_{THL2}	54	3 V	IIH								OUT	IIH	[Connect to 2B				10			

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Note	Measured terminal	Test limits		Unit
			0	1A	1Y	1B	1C	1E	GND	Sub	2E	2C	2A	2Y	2A	V_{CC}			Min	Max	
10 T_A = 125°C	t_{ϕ}	155				A	B	GND		GND							See figure 13 for test	5	-	22.5	ns
	t_{ϕ}	156								-	GND	B	A				circuit and waveforms	10	-	22.5	-
	t_S	157				A	B	GND		-							-	5	-	34.5	-
	t_S	158								-	GND	B	A				-	10	-	34.5	-
	t_R	159				A	B	GND		-							-	5	-	30	-
	t_R	160								-	GND	B	A				-	10	-	30	-
	t_F	161				A	B	GND		-							-	5	-	22.5	-
	t_F	162								-	GND	B	A				-	10	-	22.5	-
	t_{PLH1}	163	3 V	IN	OUT					-						4.5 V	See figure 11 for test	2 to 3	-	45	-
	"	164	IN	3 V	OUT					-						"	circuit and waveforms	1 to 3	-	-	-
	"	165	3 V							-				OUT	IN	"	-	13 to 12	-	-	-
8	"	166	IN							-				OUT	3 V	"	-	1 to 12	-	-	-
	t_{PHL1}	167	3 V	IN	OUT					-						"	-	2 to 3	-	22.5	-
	"	168	IN	3 V	OUT					-						"	-	1 to 3	-	-	-
	"	169	3 V							-				OUT	IN	"	-	13 to 12	-	-	-
	"	170	IN							-				OUT	3 V	"	-	1 to 12	-	-	-
	t_{PLH3}	171	3 V	IN	Connect to 1B		OUT			-						"	See figure 12 for test	2 to 5	-	90	-
	"	172	IN	3 V	Connect to 1B		OUT			-						"	circuit and waveforms	1 to 5	-	-	-
	"	173	3 V							-		OUT		Connect to 2B	IN	"	-	13 to 10	-	-	-
	"	174	IN							-		OUT		Connect to 2B	3 V	"	-	1 to 10	-	-	-
	t_{PHL3}	175	3 V	IN	Connect to 1B		OUT			-						"	-	2 to 5	-	75	-
	"	176	IN	3 V	Connect to 1B		OUT			-						"	-	1 to 5	-	-	-
	"	177	3 V							-		OUT		Connect to 2B	IN	"	-	13 to 10	-	-	-
	"	178	IN							-		OUT		Connect to 2B	3 V	"	-	1 to 10	-	-	-
	t_{TLH2}	179	3 V	IN	Connect to 1B		OUT			-						"	-	5	-	30	-
	t_{TLH2}	180	3 V							-		OUT		Connect to 2B	IN	"	-	10	-	-	-
	t_{THL2}	181	3 V	IN	Connect to 1B		OUT			-						"	-	5	-	-	-
	t_{THL2}	182	3 V							-		OUT		Connect to 2B	IN	"	-	10	-	-	-

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroups	Symbol	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Note	Measured terminal	Test limits Min	Test limits Max	Unit
			O	1I	1Y	1A	1C	1E	GND	2S	2E	2C	2A	2Y	2I	VCC					
11 T_A = 25°C	t_D	183					B		GND								See figure 13 for test circuit and waveforms	5		22.5	ns
	t_D	184					B		GND				A					10		22.5	
	t_{TS}	185					B		GND									5		36.5	
	t_{TS}	186					B		GND				A					10		36.5	
	t_R	187					B		GND									5		30	
	t_R	188					B		GND				A					10		30	
	t_F	189					B		GND									5		22.5	
	t_F	190					B		GND				A					10		22.5	
	t_{PLH1}	191		1I	OUT											4.5 V	See figure 11 for test circuit and waveforms	2 to 3		45	
		192		3 V	OUT													1 to 3			
		193		3 V										OUT	1I	3 V		13 to 12			
		194		1I										OUT				1 to 12			
	t_{PHL1}	195		1I	OUT													2 to 3		22.5	
		196		3 V	OUT													1 to 3			
		197		3 V										OUT	1I			13 to 12			
		198		1I										OUT		3 V		1 to 12			
	t_{PLH3}	199		1I	Connect to 1B			OUT									See figure 12 for test circuit and waveforms	2 to 5		90	
		200		3 V	Connect to 1B			OUT										1 to 5			
		201		3 V										(Connect to 2B	1I			13 to 10			
		202		1I										(Connect to 2B	3 V			1 to 10			
	t_{PHL3}	203		1I	Connect to 1B			OUT										2 to 5		75	
		204		3 V	Connect to 1B			OUT										1 to 5			
		205		3 V										(Connect to 2B	1I			13 to 10			
		206		1I										(Connect to 2B	3 V			1 to 10			
	t_{TLH2}	207		1I	Connect to 1B			OUT										5		30	
	t_{THL2}	208		3 V	Connect to 1B			OUT						(Connect to 2B	1I			10			
	t_{THL2}	209		1I	Connect to 1B			OUT										5			
	t_{THL2}	210		3 V										(Connect to 2B	1I			10			

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TABLE III. Group A inspection for device type 07.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	14	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	16				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	17	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL5}	18	GND	GND		"		GND	GND	"	"	8	-	76	"
	V_{OL1}	19	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL1}	20				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL5}	21	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL5}	22				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH1}	23	2 V	2 V	30 V	"				"	"	3	-	300	uA
	I_{OH1}	24				"	30 V	2 V	2 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
T_A = 125°C [2]	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	38	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	40				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	41	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL8}	42	GND	GND		"		GND	GND	"	"	8	-	76	"
	V_{OL1}	43	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL1}	44				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	45	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	46				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH1}	47	2 V	2 V	30 V	"				"	"	3	-	300	uA
	I_{OH1}	48				"	30 V	2 V	2 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
3 T_A = -55°C	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"	circuit	2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	62	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	64				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	65	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL5}	66	GND	GND		"		GND	GND	"	"	8	-	76	"
	V_{OL1}	67	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL1}	68				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL5}	69	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL5}	70				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH1}	71	2 V	2 V	30 V	"				"	"	3	-	300	uA
	I_{OH1}	72				"	30 V	2 V	2 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	3 V	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	55	ns
	"	74	3 V	IN	OUT	"				"		2 to 3	-	"	"
	"	75				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	76				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	3 V	OUT	"				"	"	1 to 3	-	40	"
	"	78	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	80				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	20	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	"	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	"	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	"	"
10 T_A = 125°C	t_{PLH}	85	IN	3 V	OUT	"				"	"	1 to 3	-	65	"
	"	86	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	88				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	3 V	OUT	"				"	"	1 to 3	-	60	"
	"	90	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	92				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	26.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	26.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	3 V	OUT	GND				4.5 V	See figure 14 for test	1 to 3	-	65	ns
	"	98	3 V	IN	OUT	"				"	circuit and waveforms	2 to 3	-	"	"
	"	99				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	100				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	3 V	OUT	"				"	"	1 to 3	-	60	"
	"	102	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	104				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	26.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	26.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 08.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	14	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	16				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH3}	17	GND	GND		"		GND	GND	"	"	8	-	17	"
	I_{CCL6}	18	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	76	"
	V_{OL2}	19	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL2}	20				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL6}	21	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL6}	22				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH2}	23	0.8 V	4.5 V	30 V	"				"	"	3	-	300	uA
	I_{OH2}	24				"	30 V	0.8 V	4.5 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
2 T_A = 125°C	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"	"	2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	38	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	40				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH3}	41	GND	GND		"		GND	GND	"	"	8	-	17	"
	I_{CCL6}	42	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	76	"
	V_{OL2}	43	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL2}	44				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL6}	45	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL6}	46				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH2}	47	0.8 V	4.5 V	30 V	"				"	"	3	-	300	uA
	I_{OH2}	48				"	30 V	0.8 V	4.5 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
3 T_A = -55°C	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	5.5 V		"				"	"	1 to 4	-	-1.6	"
	"	62	5.5 V	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	5.5 V	"	"	6 to 4	-	"	"
	"	64				"		5.5 V	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH3}	65	GND	GND		"		GND	GND	"	"	8	-	17	"
	I_{CCL6}	66	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	76	"
	V_{OL2}	67	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL2}	68				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL6}	69	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL6}	70				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH2}	71	0.8 V	4.5 V	30 V	"				"	"	3	-	300	uA
	I_{OH2}	72				"	30 V	0.8 V	4.5 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	3 V	OUT	GND				4.5 V	See figure 14 for test	1 to 3	-	65	ns
	"	74	3 V	IN	OUT	"				"	circuit and waveforms	2 to 3	-	"	"
	"	75				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	76				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	3 V	OUT	"				"	"	1 to 3	-	50	"
	"	78	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	80				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	25	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	25	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	20	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	20	"
10 T_A = 125°C	t_{PLH}	85	IN	3 V	OUT	"				"	"	1 to 3	-	90	"
	"	86	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	88				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	3 V	OUT	"				"	"	1 to 3	-	75	"
	"	90	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	92				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	33.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	33.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	3 V	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	90	ns
	"	98	3 V	IN	OUT	"	OUT	3 V	IN	"	"	2 to 3	-	"	"
	"	99				"		IN	3 V	"	"	6 to 5	-	"	"
	"	100				"	OUT	3 V	IN	"	"	7 to 5	-	75	"
	t_{PHL}	101	IN	3 V	OUT	"				"	"	1 to 3	-	"	"
	"	102	3 V	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	3 V	"	"	6 to 5	-	"	"
	"	104				"	OUT	3 V	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	33.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	33.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 09.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	14	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	16				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	17	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL5}	18	GND	GND		"		GND	GND	"	"	8	-	76	"
	V_{OL3}	19	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL3}	20				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL7}	21	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL7}	22				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH3}	23	2 V	GND	30 V	"				"	"	3	-	300	uA
	I_{OH3}	24				"	30 V	2 V	GND	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 09 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
2 T_A = 125°C	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	38	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	40				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	41	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL5}	42	GND	GND		"		GND	GND	"	"	8	-	76	"
	V_{OL3}	43	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL3}	44				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL7}	45	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL7}	46				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH3}	47	2 V	GND	30 V	"				"	"	3	-	300	uA
	I_{OH3}	48				"	30 V	2 V	GND	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 09 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
$T_A = -55°C$ [3]	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"	"	2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	62	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	64				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH2}	65	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	11	"
	I_{CCL5}	66	GND	GND		"		GND	GND	"	"	8	-	76	"
	V_{OL3}	67	0.8 V	0.8 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL3}	68				"	100 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.5	"
	V_{OL7}	69	0.8 V	0.8 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL7}	70				"	300 mA	0.8 V	0.8 V	"	"	5 to 4	-	0.8	"
	I_{OH3}	71	2 V	GND	30 V	"				"	"	3	-	300	uA
	I_{OH3}	72				"	30 V	2 V	GND	"	"	5	-	300	uA

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TABLE II!. Group A inspection for device type 09 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	GND	OUT	GND				4.5 V	See figure 14 for test	1 to 3	-	55	ns
	"	74	GND	IN	OUT	"				"	circuit and	2 to 3	-	"	"
	"	75				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	76				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	GND	OUT	"				"	"	1 to 3	-	40	"
	"	78	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	80				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	25	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	"	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	"	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	"	"
10 T_A = 125°C	t_{PLH}	85	IN	GND	OUT	"				"	"	1 to 3	-	65	"
	"	86	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	88				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	GND	OUT	"				"	"	1 to 3	-	60	"
	"	90	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	92				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	33.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	33.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 09 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	GND	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	65	ns
	"	98	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	99				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	100				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	GND	OUT	"				"	"	1 to 3	-	60	"
	"	102	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	104				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	33.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	33.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 10.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
1 T_A = 25°C	V_{IC}	1	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	2	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	3				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	4				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	5	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	µA
	"	6	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	7				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	8				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	9	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	10	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	11				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	12				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	13	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	14	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	15				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	16				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH5}	17	GND	GND		"		GND	GND	"	"	8	-	19	"
	I_{CCL8}	18	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	85	"
	V_{OL4}	19	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL4}	20				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	21	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	22				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH4}	23	0.8 V	0.8 V	30 V	"				"	"	3	-	300	µA
	I_{OH4}	24				"	30 V	0.8 V	0.8 V	"	"	5	-	300	µA

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TABLE III. Group A inspection for device type 10 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
2 T_A = 125°C	V_{IC}	25	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	26	4.5 V	-12 mA		"				"	"	2 to 4	-	"	"
	"	27				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	28				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	29	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	30	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	31				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	32				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	33	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	34	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	35				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	36				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	37	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	38	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	39				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	40				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH5}	41	GND	GND		"		GND	GND	"	"	8	-	19	"
	I_{CCL8}	42	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	85	"
	V_{OL4}	43	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL4}	44				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	45	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	46				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH4}	47	0.8 V	0.8 V	30 V	"				"	"	3	-	300	uA
	I_{OH4}	48				"	30 V	0.8 V	0.8 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 10 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
$T_A = -55°C$ [3]	V_{IC}	49	-12 mA	4.5 V		GND				4.5 V	See figure 10 for test circuit	1 to 4	-	-1.5	V
	"	50	4.5 V	-12 mA		"				"		2 to 4	-	"	"
	"	51				"		-12 mA	4.5 V	"	"	6 to 4	-	"	"
	"	52				"		4.5 V	-12 mA	"	"	7 to 4	-	"	"
	I_{IH1}	53	2.4 V	GND		"				5.5 V	"	1 to 4	-	40	uA
	"	54	GND	2.4 V		"				"	"	2 to 4	-	"	"
	"	55				"		2.4 V	GND	"	"	6 to 4	-	"	"
	"	56				"		GND	2.4 V	"	"	7 to 4	-	"	"
	I_{IH2}	57	5.5 V	GND		"				"	"	1 to 4	-	1	mA
	"	58	GND	5.5 V		"				"	"	2 to 4	-	"	"
	"	59				"		5.5 V	GND	"	"	6 to 4	-	"	"
	"	60				"		GND	5.5 V	"	"	7 to 4	-	"	"
	I_{IL1}	61	0.4 V	GND		"				"	"	1 to 4	-	-1.6	"
	"	62	GND	0.4 V		"				"	"	2 to 4	-	"	"
	"	63				"		0.4 V	GND	"	"	6 to 4	-	"	"
	"	64				"		GND	0.4 V	"	"	7 to 4	-	"	"
	I_{CCH8}	65	GND	GND		"		GND	GND	"	"	8	-	19	"
	I_{CCL8}	66	5.5 V	5.5 V		"		5.5 V	5.5 V	"	"	8	-	85	"
	V_{OL4}	67	2 V	2 V	100 mA	"				4.5 V	"	3 to 4	-	0.5	V
	V_{OL4}	68				"	100 mA	2 V	2 V	"	"	5 to 4	-	0.5	"
	V_{OL8}	69	2 V	2 V	300 mA	"				"	"	3 to 4	-	0.8	"
	V_{OL8}	70				"	300 mA	2 V	2 V	"	"	5 to 4	-	0.8	"
	I_{OH4}	71	0.8 V	0.8 V	30 V	"				"	"	3	-	300	uA
	I_{OH4}	72				"	30 V	0.8 V	0.8 V	"	"	5	-	300	uA

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TABLE III. Group A inspection for device type 10 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
9 T_A = 25°C	t_{PLH}	73	IN	GND	OUT	GND				4.5 V	See figure 14 for test circuit and waveforms	1 to 3	-	65	ns
	"	74	GND	IN	OUT	"				"		2 to 3	-	"	"
	"	75				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	76				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	77	IN	GND	OUT	"				"	"	1 to 3	-	50	"
	"	78	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	79				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	80				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	81	IN	IN	OUT	"				"	"	3	-	20	"
	t_{TLH}	82				"	OUT	IN	IN	"	"	5	-	"	"
	t_{THL}	83	IN	IN	OUT	"				"	"	3	-	"	"
	t_{THL}	84				"	OUT	IN	IN	"	"	5	-	"	"
10 T_A = 125°C	t_{PLH}	85	IN	GND	OUT	"				"	"	1 to 3	-	90	"
	"	86	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	87				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	88				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	89	IN	GND	OUT	"				"	"	1 to 3	-	75	"
	"	90	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	91				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	92				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	93	IN	IN	OUT	"				"	"	3	-	26.5	"
	t_{TLH}	94				"	OUT	IN	IN	"	"	5	-	26.5	"
	t_{THL}	95	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	96				"	OUT	IN	IN	"	"	5	-	25	"

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TABLE III. Group A inspection for device type 10 - Continued.

Subgroup	Symbol	Test no.	1	2	3	4	5	6	7	8	Note	Measured terminal	Test limits		Unit
			1A	1B	1Y	GND	2Y	2A	2B	V_{CC}			Min	Max	
11 T_A = -55°C	t_{PLH}	97	IN	GND	OUT	GND				4.5 V	See figure 14 for test	1 to 3	-	90	ns
	"	98	GND	IN	OUT	"				"	circuit and waveforms	2 to 3	-	"	"
	"	99				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	100				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{PHL}	101	IN	GND	OUT	"				"	"	1 to 3	-	75	"
	"	102	GND	IN	OUT	"				"	"	2 to 3	-	"	"
	"	103				"	OUT	IN	GND	"	"	6 to 5	-	"	"
	"	104				"	OUT	GND	IN	"	"	7 to 5	-	"	"
	t_{TLH}	105	IN	IN	OUT	"				"	"	3	-	26.5	"
	t_{TLH}	106				"	OUT	IN	IN	"	"	5	-	26.5	"
	t_{THL}	107	IN	IN	OUT	"				"	"	3	-	25	"
	t_{THL}	108				"	OUT	IN	IN	"	"	5	-	25	"

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) For class S devices, dynamic tests test condition D, using the circuit shown on figure 4; test duration shall be 240 hours minimum. Static tests (test condition A), using the circuit shown on figure 5; test duration shall be 48 hours minimum.

(2) For class B devices, test condition A, using the circuit shown on figure 5; or test condition D, using the circuit shown on figure 4.

b. Interim and final electrical test parameters shall be as specified in table II except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 5. If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition A, using the circuit shown on figure 5; or test condition D, using the circuit shown on figure 4; or test condition F, using the circuit shown on figure 5.

(2) T_A = 125°C minimum.

(3) Test duration: 1000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Special subgroups shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits of subgroups 10 and 11 as specified in table III herein.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

TABLE IV. Group C end-point electrical parameters.
T_A = 25°C ±V_{CC} = ±4.5 V

Test	Device types 01 thru 10			
	Limits		Delta	Unit
	Min	Max		
I_{OH}	---	300	±50	µA

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the ground terminal of the device under test (DUT). Current values given are for conventional current and are positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at 25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

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d. Requirements for notification of change of product to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance provisions.

g. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{IC}	Input clamp voltage.
I_{IH}	High level input current (V_{IN} = 2.4 or 5.5 V).
I_{IL}	Low level input current (with V_{IN} = .4).
I_{CCH}	High level supply current. This is the supply current with the output high.
I_{CCL}	Low level supply current. This is the supply current with the output low.
V_{OL}	Low level output voltage. This at rated load for the TTL gates.
I_{OH}	High level output current.
V_{OH}	High level output voltage, TTL gate.
I_{OS}	Short-circuit output current TTL gate.
V_{CBO}	Collector-base breakdown voltage, separate transistor.
V_{CER}	Collector-base breakdown voltage with R_{BE} = 500 ohms.
V_{CEO}	Collector-emitter breakdown voltage.
V_{BEO}	Emitter-base breakdown voltage.
h_{FE}	Static forward current transfer ratio.
V_{BE}	Base-emitter voltage.
$V_{CE(SAT)}$	Collector-emitter saturation voltage.
t_D	Delay time.
t_R	Rise time.
t_S	Storage time.
t_F	Fall time.
t_{PLH}	Propagation delay time (low to high level output transition).
t_{PHL}	Propagation delay time (high to low level output transition).
t_{TLH}	Transition time (low to high level output transition).
t_{THL}	Transition time (high to low level output transition).

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6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	55450
02	55451
03	55452
04	55453
05	55454
06	55460
07	55461
08	55462
09	55463
10	55464

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Civil Agencies Coordinating Activities:
NASA - NA

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0791)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/129A	Microcircuits, Monolithic Silicon Interface, Dual Peripheral Drivers

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify):
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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